UAV-NOMA Disaster Network Planning Simulator

A simulation-based planning tool for UAV base stations using NOMA in disaster areas, with a fair evaluation of reinforcement-learning control

A major project report submitted in partial fulfilment of the requirements for the award of the degree of

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Under the guidance of

[PENDING: Guide name, designation, department]

[PENDING: Department of ..., College name, City]

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[PENDING: Academic year, e.g. 2026-2027]

Certificate

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Guide	Head of the Department	Principal
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Declaration

We, the students named on the title page, declare that the work presented in this report is our own, carried out under the guidance of [PENDING: guide], and has not been submitted for any other degree. Sources are cited in the References.

Disclosure of AI assistance [PENDING: confirm the wording against the university's policy]: much of the simulator, the baseline pilots and the reinforcement-learning pipeline was first written in AI-assisted sessions (an AI coding assistant) with the project owner, and the team then took ownership to review, explain and extend the code; every result was re-run, checked and recorded in the project book.

[PENDING: place, date, signatures]

Acknowledgement

[PENDING: thanks to the guide, the head of department, the principal, the institution, family and friends.] The idea of a moving base station using NOMA came from our guide; the team extended it with moving crowds, collapsed buildings, a battery limit and a learning controller.

Abstract

When an earthquake or a flood destroys the mobile towers in an area, a drone carrying a small base station can restore service for a while, but where it should fly, how high, and how it should share its radio resources are open questions. We built an open, tested, simulation-based planning tool for this problem: moving survivor crowds with five service classes, collapsed buildings, the Al-Hourani air-to-ground channel, the Zeng-Xu-Zhang rotary-wing energy model with a battery, and a NOMA downlink in which two users share one frequency slice. Every strategy flies the same seeded scenarios and is compared with paired 95% confidence intervals, with pass marks written down before the results. Results: a drone that follows the crowd serves 74 of every 100 people where a parked drone serves 43; when NOMA and OMA are optimised in the same way, adaptive NOMA serves 3 to 5 more people per 100 for every pilot, and about 13 more for the best pilot in a scenario based on the 2024 Kathmandu floods with real OpenStreetMap buildings; circling at the minimum-power speed serves about 26% more people per battery than hovering. A planner keeps 98% of its coverage when only half of the phones report a GPS position (one in twenty on competing sites), even with 50 m error. A "clairvoyant" pilot that knows the future measures, before any training, whether an AI has room to win at all: on walking crowds there is almost none (+0.002); room exists only where demand jumps between sites about 2 km apart. A rule meant to predict that room from distance, speed and timing was not confirmed on new designs. The learning pilots did not beat the best simple pilot anywhere: not on still crowds (0.576 against a pass mark of 0.580), not on competing sites (best 0.680 against 0.756), not as a hybrid that only decides when to leave (-0.004 and -0.019), and not when the true radio channel differs from the textbook model (-0.039 to -0.101). A simulated quadcopter with an autopilot flew the same plans within 0.012 coverage of the simplified game. The result is an honest map of where learning helps and where a computed rule is enough.

Keywords: UAV base station, NOMA, disaster communication, air-to-ground channel, rotary-wing energy, reinforcement learning, baselines, simulation-based planning.

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Chapter 1. Introduction

1.1 Problem statement

After an earthquake, a cyclone or a flood, the ground mobile network in the affected area often fails: towers lose power, backhaul cables break and buildings collapse. Survivors cannot call for help and rescue teams cannot coordinate. A drone (unmanned aerial vehicle, UAV) carrying a small base station can act as a temporary tower. Three things make this hard:

1. the drone's battery lasts minutes, not hours, and flying (not the radio) dominates its energy use;
2. the air-to-ground signal depends on the drone's height and position, and collapsed buildings block some links;
3. survivors move, sometimes suddenly, and one drone must share its bandwidth and power among hundreds of people with different needs.

The planning question is therefore: **where should the drone fly, how high, and how should it share its radio resources over time so that as many survivors as possible stay connected before the battery runs out?**

1.2 Motivation

- Restoring a basic connection quickly matters more in a disaster than giving already-served users a fast one. This defines our main metric, **coverage**: the share of users who receive at least the minimum data rate of their class.
- Non-orthogonal multiple access (**NOMA**) lets two users share the same frequency slice at the same time by splitting power, so a power- and bandwidth-limited drone can serve more people. The guide's original idea was a moving base station using NOMA.
- Rescue planners cannot rehearse drone strategies on real disasters. A **simulation-based planning tool** lets strategies be compared fairly in software before any flight, saving hardware, battery and flight time. We deliberately do not call it a digital twin: that term needs live or replayed state data, which this project does not have.
- Reinforcement learning (RL) is often proposed for UAV control. We wanted to know honestly **when** a learned controller beats strong non-learning strategies and when it does not.

1.3 Objectives

1. Build an open, tested, reproducible simulator of a UAV base station serving moving survivor crowds with a NOMA downlink, with every physical parameter sourced or labelled as an engineering assumption.
2. Build strong non-learning pilots (baselines) and a fair evaluation protocol (same scenarios, paired statistics, fixed train/validation/test seeds, pass marks written down before results).
3. Measure NOMA against OMA inside the mission, for every pilot, including an equally optimised OMA.
4. Train RL pilots that learn by themselves from raw measurements and report where they win and lose.
5. Measure the room any anticipating controller could have (a clairvoyant upper bound) before spending effort on training.
6. Check that the simplified flight model does not distort the conclusions, by flying the same plans with a simulated quadcopter and autopilot.

7. Provide a dashboard so that non-specialists can set up a disaster, fly any pilot and compare them.
8. Measure how much phone position information a planner needs, and test whether learning helps when the real radio channel differs from the textbook model.
9. Run the pilots on a scenario based on a real place (real buildings, reconstructed people) to check that the conclusions survive real geometry.

Wording rule. "Simulation-based planning tool", never "digital twin": the project has no live state synchronisation and no replayed real traces. A scenario built on a real map is called "a scenario based on ..." and its people are never called real data.

1.4 Hypotheses

Every experiment serves one of these hypotheses.

Table 1.1 Hypotheses

ID	Hypothesis	In plain words
H1	Adaptive UAV control beats a fixed UAV once crowds move	A drone that follows people does better than one parked in the middle
H2	RL beats strong heuristics (k-means, grid search, particle swarm) only when dynamics and battery limits make them short-sighted; report where it wins and where it does not	The AI should only win in hard, changing situations, and we must say where it loses
H3	NOMA pairing improves throughput and coverage over OMA at equal bandwidth and power	Sharing a slice between two people beats giving everyone a separate slice
H4 (stretch)	Coordinated multi-UAV control beats independent placement	Several drones working together beat drones acting alone

1.5 Scope

In scope: one UAV; 2 x 2 km and 1 x 1 km areas; 100-1,000 users in 2-5 crowds; six crowd-movement models and competing-site events; collapsed buildings as 2D rectangles; the Al-Hourani mean path loss with optional fading; the Zeng-Xu-Zhang energy model with a battery; NOMA (fixed and adaptive) and OMA downlinks; non-learning pilots (B0-B8, B6fc, B6dep and diagnostic variants), PPO and SAC agents, a hybrid timing agent; a clairvoyant planner; a simulated quadcopter with an autopilot; a Streamlit dashboard; one case study on real building footprints (a 2 x 2 km area of Kathmandu from OpenStreetMap, with a height-aware line-of-sight test), with the people reconstructed from assumptions.

Out of scope: real hardware and flight tests; terrain, and building heights outside the one real-map case; the drone's own backhaul; interference from other cells; bit-error rate; locating survivors from their phones (future work, Section 7.3); more than one drone (H4 was not attempted). All results are conditional on the model: they may be used to compare strategies under the stated assumptions, not to predict absolute coverage in a real disaster.

1.6 Organisation of the report

Chapter 2 surveys the literature by theme. Chapter 3 describes the system design. Chapter 4 covers the implementation, testing and reproducibility. Chapter 5 presents the results, hypothesis by hypothesis, then the studies of location information and the Kathmandu case. Chapter 6 discusses where learning helps, the limitations and threats to validity. Chapter 7 concludes, lists future work and describes the next study, finding survivors from their phones.

Chapter 2. Literature survey

The survey is organised by the six themes the project touches. The full matrix with links, and what we took from each paper, is in `docs/report/literature.md`. **Every entry still needs a person to check it against the original** (marked VERIFY there); of the two papers used for parameters, only [2] was read in full.

2.1 UAV base-station placement

A UAV base station's coverage depends on its 3D position. Bor-Yaliniz, El-Keyi and Yanikomeroglu [5] formulate 3D placement of one aerial base station as a mixed-integer non-linear problem and give an efficient numerical solution. Alzenad et al. [6] place a UAV base station to cover the most users with the least transmit power. Lyu, Zeng, Zhang and Lim [7] place several UAV-mounted base stations along a spiral to cover all ground terminals with as few drones as possible. Mozaffari et al. [4] survey placement, trajectory, energy and channel modelling for UAVs in wireless networks. These works optimise placement for a known, mostly static user set; our search pilots B3, B4 and B6 are simple members of this family (grid search, particle swarm [31]), used as strong baselines rather than as contributions.

2.2 The air-to-ground channel

Al-Hourani, Kandeepan and Lardner [1] give a closed-form probability of line of sight between a low-altitude platform and a ground user as a function of the elevation angle, and show that an optimal altitude exists for maximum coverage. The mean path loss is the free-space loss plus an excess loss weighted by the line-of-sight probability. We use this model with a = 9.61, b = 0.16, eta_LoS = 1 dB and eta_NLoS = 20 dB at 2 GHz. **The original letter was not read** (paywalled); the formulas and parameter values were checked in two secondary papers that cite it [27], [28], which list them without naming the environment, so the label "urban" is unconfirmed.

2.3 Rotary-wing energy

Zeng and Zhang [3] model the propulsion energy of a fixed-wing UAV and propose a circular trajectory with optimised radius and speed for energy efficiency. Zeng, Xu and Zhang [2] derive the propulsion power of a rotary-wing UAV as a function of speed: blade profile power, induced power and parasite power. Its key consequence for us is that moderate forward flight costs less than hovering. We verified the formula and constants (delta, k, rho, s) in [2] and derived our own small 20 N drone from it (hover 168.49 W); the fuselage drag ratio d0 = 0.6 is unverified ([2] uses 0.3).

2.4 NOMA and UAV-NOMA

Saito et al. [8] proposed power-domain NOMA with successive interference cancellation (SIC) for cellular radio access. Ding, Fan and Poor [9] show that with fixed power allocation, pairing users with very different channels enlarges NOMA's gain over OMA, and they study cognitive-radio-inspired NOMA in which the weak user's rate is protected. Liu et al. [10] present a framework for UAV communications based on NOMA. Cui et al. [11] jointly design the UAV trajectory and resource allocation for OMA and NOMA under max-min fairness. Chen et al. [12] allocate power in a NOMA-UAV network flying a circular trajectory, with a secrecy constraint. Our fixed scheme (strongest paired with weakest, 0.8/0.2 split) follows the fixed-allocation idea of [9]; our adaptive scheme gives each pair its minimum power in closed form, which is close in spirit to the rate-protecting allocation of [9].

2.5 Reinforcement learning for UAV networks

Deep RL has been applied to UAV base stations many times. Ghanavi et al. [16] use Q-learning to place an aerial base station for moving users. Liu, Liu, Chen and Hanzo [15] design trajectories and power control for multi-UAV networks with machine learning. Zhong et al. [13] combine

k-means with a deep Q-network to control the 3D trajectory and power of a UAV offloading NOMA users. AdaptSky [14] uses deep RL for joint NOMA power allocation and UAV placement. Hu et al. [17] use meta-RL so that an energy-limited UAV adapts its trajectory quickly to new demand. Rizvi and Boyle [18] use multi-agent RL with action masking for UAV mobile communications, and Wang et al. [19] split beamforming (fast, graph network) from trajectory (slow, multi-agent RL) in a two-layer design. **So the method is not new: we cannot claim to be the first to apply RL to UAV-NOMA.** In the papers we looked at, RL is usually compared with random, circular or OMA references; comparisons against strong search pilots on the same scenarios with paired statistics were rarer (to be checked across the full matrix).

The learners we use are PPO [20] and SAC [21] from Stable-Baselines3 [23], in a Gymnasium environment [24]. The privileged ("asymmetric") critic that sees the true state during training only follows Pinto et al. [22].

2.6 Evaluation practice

Henderson et al. [25] showed that deep-RL results vary strongly with random seeds and implementation details, and that few seeds can produce misleading conclusions. Agarwal et al. [26] recommend interval estimates and robust aggregate metrics rather than point scores. We follow both: fixed seed splits, several training runs, paired confidence intervals, a Holm correction [29] where many comparisons are made, Jain's fairness index [30] as a secondary metric, and pass marks written down before results are seen.

2.7 Literature matrix

Table 2.1 Literature matrix (short form; details and links in literature.md; every entry: VERIFY)

Ref.	Theme	Main idea	What we take or compare
[1] Al-Hourani 2014	Channel	LoS probability vs elevation; optimal altitude	Our channel model and parameters
[2] Zeng, Xu, Zhang 2019	Energy	Rotary-wing propulsion power	Our energy model; forward flight cheaper than hover
[3] Zeng, Zhang 2017	Energy	Fixed-wing energy; circular trajectory	Motivation for the circling pilot B7
[4] Mozaffari 2019	Survey	UAVs in wireless networks	Problem framing
[5] Bor-Yaliniz 2016	Placement	3D placement as an optimisation	Search pilots as baselines
[6] Alzenad 2017	Placement	Energy-efficient maximal coverage placement	Coverage as the objective
[7] Lyu 2017	Placement	Spiral placement of several UAV base stations	Multi-UAV future work (H4)
[8] Saito 2013	NOMA	Power-domain NOMA with SIC	NOMA principle
[9] Ding, Fan, Poor 2016	NOMA	User pairing; fixed vs rate-protecting power	Extreme pairing; adaptive split
[10] Liu 2019	UAV-NOMA	Framework for NOMA-based UAV networks	Context for H3
[11] Cui 2019	UAV-NOMA	Joint trajectory and resource allocation, OMA vs NOMA	Fair OMA reference
[12] Chen 2020	UAV-NOMA	Power allocation on a circular trajectory	Circling with NOMA
[13] Zhong 2020	RL + NOMA	k-means + DQN for trajectory and power	Shows RL-for-UAV-NOMA is published
[14] Benfaid 2021	RL + NOMA	DRL for NOMA power and placement	Same
[15] Liu 2018	RL	Multi-UAV trajectory and power control by ML	Same
[16] Ghanavi 2018	RL	Q-learning placement for moving users	Closest to our G5 task
[17] Hu 2020	RL	Meta-RL trajectories for an energy-limited UAV	Adapting to new demand
[18] Rizvi, Boyle 2023	RL	Multi-agent RL with action masking	Action design; H4
[19] Wang 2026	RL	Two-layer, two-timescale RL for multi-UAV	Supports our time-scale split
[20] Schulman 2017	Method	PPO	Our main learner
[21] Haarnoja 2018	Method	SAC	Our second learner
[22] Pinto 2017	Method	Asymmetric actor-critic	Privileged critic (training only)
[23] Raffin 2021	Tool	Stable-Baselines3	Implementation of PPO and SAC
[25] Henderson 2018	Evaluation	Seed variance in deep RL	Several runs per version
[26] Agarwal 2021	Evaluation	Interval estimates for RL results	Confidence intervals, pass marks

2.8 Gap and positioning

From the survey, the defensible contribution is not a new algorithm but **an open, tested simulator and evaluation protocol for UAV-NOMA planning in disaster scenarios, a comparison of learning against strong non-learning pilots across regimes, and an analysis of when learning helps**, together with specific findings:

1. **H1:** an adaptive drone beats a parked one where the area is large and power scarce (0.741 vs 0.428), and most of a search pilot's lead can come from choosing the altitude.
2. **H3:** NOMA's gain measured inside the mission for every pilot, against an equally optimised OMA (+0.03 to +0.05), and about twice as large for the best pilot on real building geometry (Kathmandu case, +0.127).
3. **Energy:** circling at the minimum-power speed serves about 26% more people per battery than hovering.

4. **Clairvoyant headroom:** a method for deciding, before any training, whether a scenario leaves any room for a learning controller at all.
5. **Value of location information:** a design rule for how many phones must report their position, how accurately and how often (pre-registered, passed).
6. **Sim-to-sim:** the coverage conclusions survive a realistic quadcopter with an autopilot and wind.
7. **An honest map of where AI does and does not help**, including four pre-registered negative results and one rule that was not confirmed (the anticipation rule).

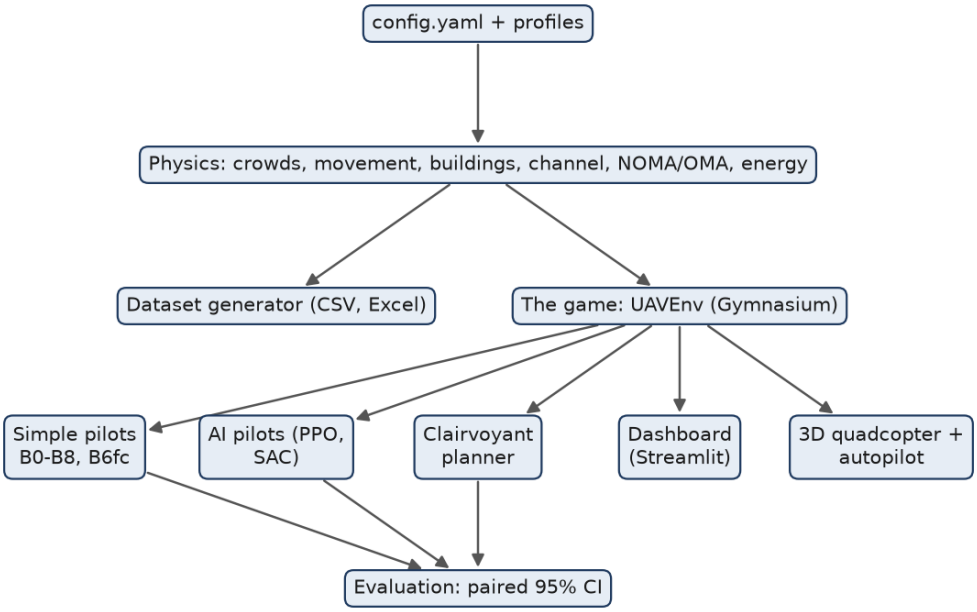
Drone systems that locate phones from the air for search and rescue have been described in a paper and in industry reports ([33], [35]; both still to be verified against the originals). Our project does not locate phones; it plans a drone that serves them. Joining the two ("find-and-serve") is the next study (Section 7.3).

Chapter 3. System design

3.1 Architecture

The design principle is **one physics code path**: the dataset generator, the game (Gymnasium environment), every pilot, the AI, the dashboard and the 3D quadcopter all call the same functions in `make_dataset.py`, so they cannot disagree. Everything below the game is plain Python and NumPy with no user-interface or RL dependency, and every setting lives in `config.yaml`.

Figure 3.1 System architecture: one physics engine, one game, many pilots



3.2 Scenarios

A scenario is one invented disaster area identified by a seed; the same seed always gives the same crowd, buildings and movement. Each seed draws the number of users (100, 300, 600 or 1,000), 2-5 clusters with radius 50-200 m, 5-10% scattered users, the movement model and the buildings. Seeds are split and never mixed: **train 0-99, validation 100-119, test 120-139**; the RL agents later trained on fresh seeds 10,000-19,999. Named profiles change only what they name.

Table 3.1 Scenario profiles

Profile	Changes from the default (1 x 1 km, 30 dBm, 0-5 buildings, five movement models, 10 Wh)
`lowpower`	20 dBm
`bigarea`	2 x 2 km
`hard`	2 x 2 km, 20 dBm, 3-8 buildings of 60-200 m, only hotspot jump / surge / group drift / attraction, 16 Wh
`rl_static`	as `hard` but the crowd stands still (RL curriculum step 1)
`sites_d{10,15,20}_t{0,60,180}`	2-3 sites 1.0-2.0 km apart; 50-70% of the demand switches from site A to B at 60-150 s, at once or over 1 or 3 minutes; G5 settings inside
`sites_v*`, `sites_T600_*`, `sites_d175_t30`, `sites_d125_t120`	Eight new competing-sites designs for testing the anticipation rule: drone top speed 8 or 25 m/s, a 600 s mission, or new distances and switch times
`mm_*`	`rl_static` with G5 settings, and a TRUE radio world that differs from the planners' textbook model: correlated shadowing, a 25 dB building loss (planners assume 15 dB) and buildings missing from the planners' map
`place_kathmandu_bagmati`	A 2 x 2 km area of Kathmandu along the Bagmati river: 12,061 real building footprints from OpenStreetMap, height-aware line of sight, a reconstructed flood crowd that does not walk, an evacuation event; 20 dBm, 16 Wh, G5 settings

Every person belongs to a class that sets the minimum rate needed to count as served.

Table 3.2 User classes and minimum rates (engineering assumptions, not yet signed off)

Class	Share	Minimum rate
rescue	10%	200 kbps
video	20%	500 kbps
voice	30%	50 kbps
iot	20%	10 kbps
normal	20%	100 kbps

Movement. Six models move every person each second for 300 s, vectorised over users: static; random waypoint (0.5-1.5 m/s); group drift; attraction to 1-2 shelters; hotspot jump (one crowd relocates at 3 m/s at a random moment between 60 and 180 s); surge (everyone runs to one point at 3 m/s). In the competing-sites profiles an activity mask switches users off at site A and equally many on at site B, so crowd size, class mix and bandwidth stay the same.

Collapsed buildings. Axis-aligned rectangles; a drone-user link whose 2D segment crosses one gets 15 dB extra loss (engineering assumption, range 10-25 dB), tested with the vectorised slab (Liang-Barsky) method.

3.3 The radio channel

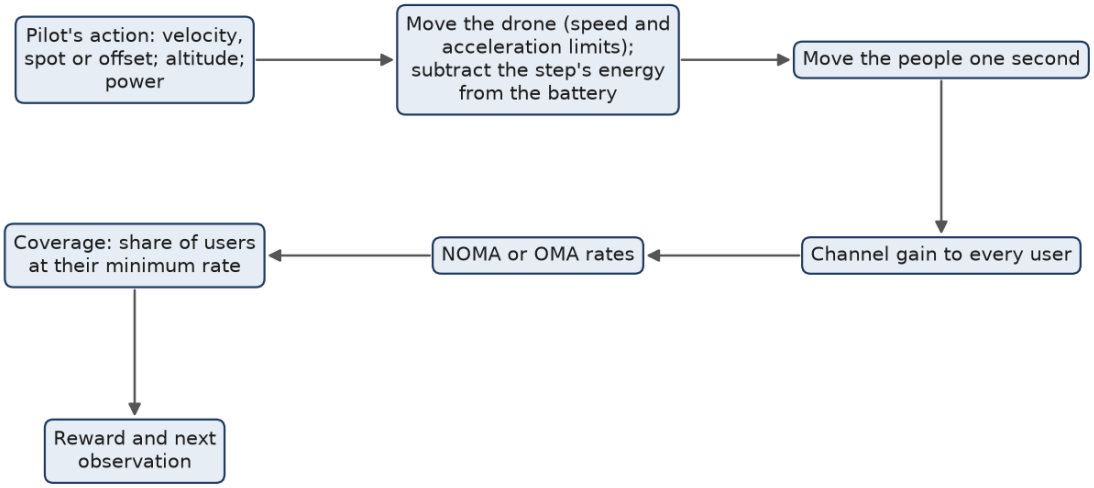
For a drone at (X, Y, H) and a user at (x, y), following Al-Hourani et al. [1]:

```
d      = sqrt((x - X)^2 + (y - Y)^2 + H^2)
theta  = (180 / pi) * arcsin(H / d)                          elevation angle in degrees
P_LoS  = 1 / (1 + a exp(-b (theta - a)))                      a = 9.61, b = 0.16
PL_dB  = FSPL(d, 2 GHz) + P_LoS * 1 dB + (1 - P_LoS) * 20 dB   (+ 15 dB if a building blocks the link)
noise  = -174 dBm/Hz + 7 dB noise figure, over each user's bandwidth
```

Optional fading (Rician K = 10 dB for LoS, Rayleigh for NLoS) is off for the dataset and the baselines. Bandwidth scales with the crowd: 50 kHz per user.

On a real map (Kathmandu case only). Al-Hourani's P_LoS is a statistical stand-in for buildings whose positions are unknown. Where the buildings are known, the simulator instead draws the straight 3D line from each phone (1.5 m above the ground or the roof it stands on) to the drone over a 2 m height grid of the footprints: a clear ray gets FSPL + 1 dB, a ray that passes below a roof gets FSPL + 20 dB (the same two Al-Hourani terms). Using both P_LoS and the known buildings would count the buildings twice; that variant is kept as a sensitivity run. No diffraction, trees, terrain or indoor users.

Figure 3.2 One second of the game



3.4 NOMA and OMA

Plain words. OMA gives every person a separate slice of the bandwidth. NOMA puts two people on the same slice: the weak user gets most of the power and reads its own message treating the other as noise; the strong user first decodes and removes the weak user's message (SIC) and then reads its own.

Four access schemes, all with the same total bandwidth B and power P (N users):

1. `oma`: FDMA, each user gets B/N and P/N.
2. `noma` (fixed): users sorted by channel gain, strongest paired with weakest; each pair gets 2B/N and 2P/N, split 0.8 to the weak user and 0.2 to the strong user (formulas below).
3. `oma_opt`: each user gets the minimum power for its rate, p_i = (2^(R_i/b) - 1) N0 b / g_i; users are admitted cheapest first while the total stays within P; leftover power is scaled up evenly.
4. `noma_opt` (adaptive NOMA): the same cheapest-first admission, then the admitted users are paired strongest with weakest on a double slice and each pair gets its minimum NOMA power in closed form, with an imperfect-SIC residual eps (formulas below). If NOMA would cost more than serving the pair separately (possible only with eps > 0), the pair uses OMA inside its slot. The largest number of users that fits is found by binary search; options `sic_eps` (0 or 0.05) and `rescue_first`.

```
fixed NOMA:     SINR_weak   = 0.8 p g_w / (0.2 p g_w + N0)
                SINR_strong = 0.2 p g_s / N0                  (after SIC)
                R = b log2(1 + SINR)
adaptive NOMA:  p_s = gamma_s N (eps gamma_w / g_w + 1 / g_s) / (1 - eps gamma_s gamma_w)
                p_w = gamma_w (p_s + N / g_w),      gamma = 2^(R / 2b) - 1
```

From G5 on, the project uses `noma_opt` with eps = 0.05 and rescue users admitted first, and always reports `oma_opt` next to it so that H3 stays a fair test.

3.5 Energy and battery

Rotary-wing propulsion power after Zeng, Xu and Zhang [2]:

```
P(V) = P0 (1 + 3 V^2 / Utip^2) + Pi sqrt( sqrt(1 + V^4 / (4 v0^4)) - V^2 / (2 v0^2) ) + 0.5 d0 rho s A V^3
hover P(0) = P0 + Pi = 79.86 + 88.63 = 168.49 W
energy per 1 s step = P(V) + W max(climb rate, 0) + P_tx / 0.4 + 5 W
```

For our 20 N drone: Utip = 120 m/s, v0 = 4.03 m/s, d0 = 0.6, rho = 1.225 kg/m^3, s = 0.05, A = 0.503 m^2. Flying at about 10 m/s costs 133.5 J/s against 176.0 J/s hovering (30 dBm, including

radio and electronics). An acceleration limit (4 m/s^2 horizontal) and a load-factor term for turning and braking (our own extension, labelled as such) close a "shaking" exploit in which a drone could stay in place at forward-flight power.

3.6 The game (Gymnasium environment)

`UAVEnv` plays one scenario second by second for 300 s or until the battery is empty. The drone starts at the map centre at 150 m, flies at most 15 m/s horizontally and 5 m/s vertically, between 50 and 300 m. The reward is the coverage of the second; each reward part is logged separately so that reward hacking can be spotted. The observation contains only what a real drone could measure: its position, velocity, battery and time, its previous coverage, throughput and fairness, and coarse 10 x 10 maps of where users and rescue users are. Later versions added an uncapped map, the number of connected users, the drone's own history and, in the competing-sites cells, the operator's announcement of the event. Pilots can control velocity (`xy`), fly to a spot (`goto`), or move by an offset (`moveby`, 0 = stay), and the AI decides every 10 s.

The rule about the AI (owner's decision): the AI learns by itself from raw measurements and the coverage score. No hints, no menu of candidate spots, no imitation of the baselines, no privileged information when flying. A critic that sees the true state during training only was allowed later and is disclosed wherever used.

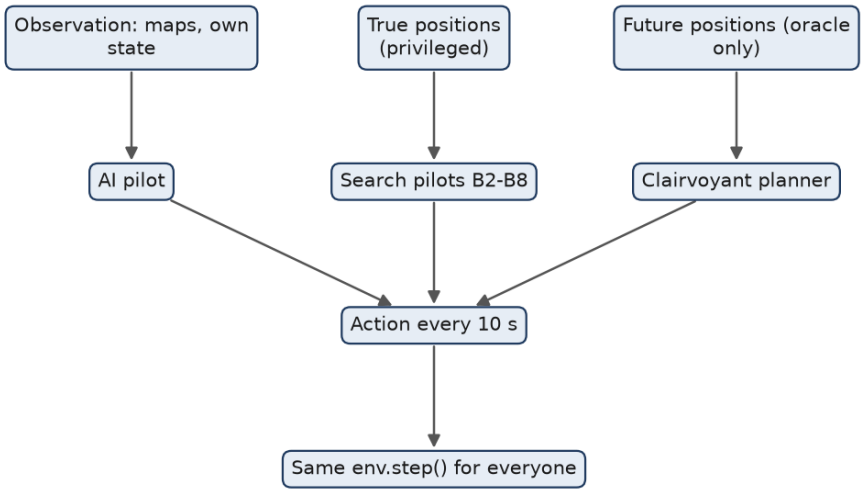
3.7 Pilots

The baselines see the true user positions (privileged information); the AI sees only maps. That makes them a strong test.

Table 3.3 The pilots

ID	Name	What it does
B0	Fixed	Parks at the map centre at 150 m, full power
B1	Follow the crowd	Flies toward the centre of the biggest true cluster
B2	Weighted k-means	Every 10 s: k = 3 clusters (rescue users count 3x), flies to the heaviest
B3	Grid search	Every 10 s: scores a grid of spots at 150 m, flies to the best
B4	Particle swarm (PSO)	Every 10 s: searches position, altitude and power
B5	Energy-aware	Like B3 but moves only if the gain pays for the energy; behaves like B3 because moving is cheaper than hovering
B6	Grid search + altitude	Like B3 with five altitudes per spot
B7	Circling	Chooses spots like B6 but circles them (50 m) at the minimum-power speed
B8	Battery rule	Like B6; circles once the battery would not last
B6fc	Forecasting	Like B6, also using the announced event schedule (competing sites)
B6dep	Departure time	B6 plus *when* to leave for the announced site: every 10 s it compares "stay until time d, then fly" for every d, with B6fc's forecast
B6rep, B3rep	Phone reports	B6's / B3's search on positions reported by a share of the phones, with GPS error and age (value of information)
B6meas	Measure and search	Textbook model to pick a few spots, then flies there, measures its real coverage and searches locally (model mismatch)
B6true	Upper bound (mismatch)	B6 with a finer search on the TRUE channel; not a competitor
B3obs, B3obsU, B3obsUO	Information ladder	B3's search using only the AI's maps (diagnostic)
Clairvoyant	Upper bound	Knows the crowd's whole future; dynamic programming over B6's grid
AI	PPO / SAC	Neural network trained by RL; sees only raw measurements
Hybrid	B6 + PPO	B6 decides where; a small PPO policy decides only when to leave for the announced site (disclosed as a hybrid)

Figure 3.3 How each pilot reaches the game



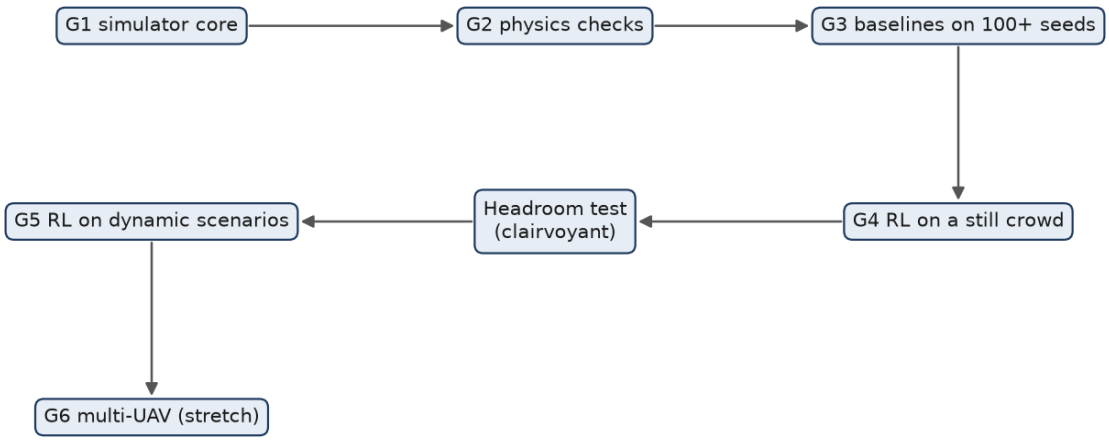
3.8 The AI

PPO [20] (and SAC [21] for comparison) from Stable-Baselines3 [23]. Two network types: a two-layer MLP (64 units) and a small CNN that reads the maps as a picture with coordinate channels. The best still-crowd version (E0) uses a dictionary observation with uncapped maps, the crowd size and the last 4 positions and coverages, fresh training scenarios and a planning horizon gamma = 0.95 matched to 30 decisions per mission. Training runs on a laptop CPU (about 1,000 simulated seconds per second when plugged in) and, for G5, on Kaggle.

3.9 The evaluation protocol

- Main metric **coverage_full**: coverage averaged over the 300 s mission, counting zero after the battery dies.
- Mean and 95% confidence interval over scenarios; **paired differences** per scenario (pilot A minus pilot B on the same scenario) so that "this scenario was easy for everyone" cancels; a * marks a CI that excludes zero.
- Holm correction where a table makes many comparisons.
- Pass marks fixed **before** results are seen (G4: D-43; competing sites: D-52; G5: D-55); test seeds read once at the end.
- Gates: a layer starts only after the previous gate passes.

Figure 3.4 The evaluation protocol and the gates

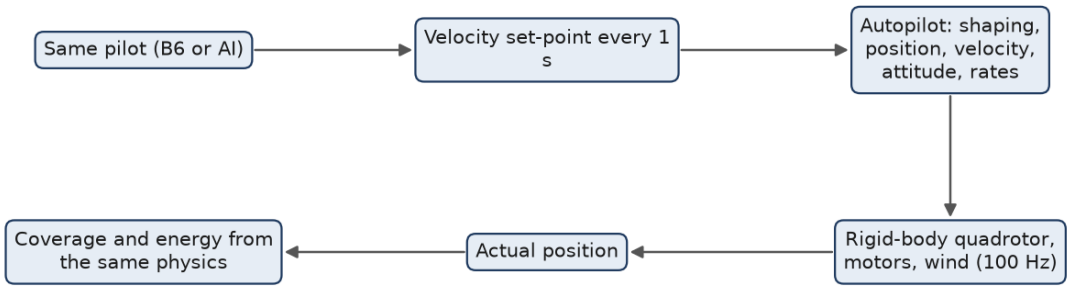


3.10 The dashboard and the 3D quadcopter

Dashboard. A Streamlit web page on the same engine, with pages to see the showcase, fly a disaster with any pilot, compare pilots, get a plan with battery advice, and (since Stage 5) watch a 3D flight. Its numbers equal the experiments' because it calls the same code; 16 tests check this without Streamlit.

Simulated quadcopter (Stage 5). In the game the drone moves like a chess piece. In Stage 5 every pilot's velocity command instead becomes the set-point of an autopilot (like ArduPilot GUIDED velocity mode) flying a rigid-body quadrotor model of the same 2.04 kg drone: thrust and attitude dynamics, a first-order motor lag, rotor and fuselage drag, a 30 degree tilt limit, a thrust-to-weight ratio of 2, a geofence, and wind with seeded gusts, integrated at 100 Hz. Energy is still computed by the Zeng model, from the airspeed, climb and acceleration of every integration step. The autopilot values quoted from ArduPilot and PX4 are marked as not verified.

Figure 3.5 The 3D quadcopter in the loop (sim-to-sim)



Chapter 4. Implementation

4.1 Tools

Python with NumPy, pandas, matplotlib, PyYAML, Gymnasium, pytest, xlsxwriter and openpyxl for the simulator and the baselines; Stable-Baselines3, PyTorch (CPU) and TensorBoard for the AI; Streamlit for the dashboard; ReportLab for the PDF documents. Development on Windows 11 (laptop CPU); continuous integration on GitHub with Python 3.11 and 3.12.

4.2 Code structure

Table 4.1 Code structure (`uav_dataset/`)

File	Role
`config.yaml`	Every setting and profile; values are changed here, never in the code
`make_dataset.py`	Physics, scenarios, movement, buildings, channel, NOMA/OMA access schemes, energy, dataset tables
`uav_env.py`	The Gymnasium game
`baselines.py`	Pilots B0-B8, B6fc and the information-ladder pilots
`run_baselines.py`, `evalstats.py`	Runner and statistics (mean +/- 95% CI, paired differences)
`rl_utils.py`, `rl_net.py`, `rl_train.py`, `rl_eval.py`	Control modes, decision hold, CNN, training and evaluation
`oracle.py`	Clairvoyant planner (dynamic programming)
`sites_exp.py`, `g5_eval.py`	Competing-sites experiments and the G5 evaluation
`noma_vs_oma.py`, `noma_picture.py`, `access_benchmark.py`	H3 experiments and pictures
`accel_loiter_exp.py`	Circling experiment
`quadsim.py`, `sim2sim_exp.py`, `viewer3d.html`	Simulated quadcopter, sim-to-sim check, 3D view
`dashboard.py`, `dashboard_core.py`	Streamlit dashboard and its tested logic
`view_run.py`, `compare_run.py`	Single-pilot GIFs and multi-pilot comparison pictures
`osm_place.py`, `place.py`, `nepal_case.py`	Real-map case: OpenStreetMap import, height grid and 3D line of sight, Kathmandu runs
`voi_exp.py`	Value of location information (study branch)
`anticip_exp.py`, `anticip_report.py`	Anticipation rule (study branch)
`hybrid.py`, `hybrid_tables.py`, `hybrid_train.py`, `hybrid_eval.py`	Hybrid timing AI (study branch)
`mismatch_exp.py`, `mismatch_eval.py`	Model mismatch (study branch)
`test_*.py`	Automated tests

4.3 Testing and continuous integration

- **399 automated tests** (2026-10-23), covering the NOMA maths (including brute force on small crowds and an independent re-derivation of the closed form, exact to 1e-15 over 2,000 random pairs), the mobility models, the obstacle geometry, the energy model, the Gymnasium checker, every pilot, the statistics, the clairvoyant planner (equal to brute force on a small grid), the privileged-critic guard (the flying network's output is bit-identical when only the privileged data changes), the dashboard logic and the quadcopter (26 tests). The later studies add their own test files (`test_place.py` on main; `test_voi.py`, `test_anticip.py`, `test_b6dep.py`, `test_hybrid.py`, `test_mismatch.py` on their branches), for example: the report-based planner with full information equals B6 action for action, the 3D ray test equals brute force, and the hybrid pilot that never leaves reproduces B6 exactly. [PENDING: total test count after the study branches are merged.]
- Every pull request runs ruff lint, pytest on Python 3.11 and 3.12 with an 80% coverage gate on the core modules, a smoke run of the generator, and a separate RL job with a short PPO training.

- `main` is protected: pull requests, green CI and code-owner review.
- Bugs caught this way include the fading flag not reaching the channel, a missing dependency, a policy adapter used before `reset()`, and a "shaking" energy exploit found by a reviewer and closed with an acceleration limit.

4.4 Reproducibility

- Seeded random generators only; the same seed gives the same scenario; a training run killed by a session restart was re-run and printed the same validation scores.
- New features are off by default, so every earlier number stays bit-exact (tested).
- Generated data is not committed but regenerated in about a minute from `config.yaml` and the seeds.
- Every model saves its settings next to it; results come only from committed configs and seeds.
- Every physical parameter has a value, unit, source or "engineering assumption", and a sensitivity range (`docs/assumptions.md`).
- A project book records every step, decision and result, with corrections added as new entries rather than rewrites.

Table 4.2 Dataset tables (version 3, 140 scenarios)

Table	Rows	One row is
scenarios	140	one scenario
obstacles	341	one collapsed building
users	74,900	one person at t = 0
trajectory	823,900	one person at one 30 s snapshot
sweep	813,120	one drone position, altitude and power in one scenario

Chapter 5. Results

All numbers are coverage (share of users served) unless stated; values are means over scenarios with 95% confidence intervals where given; * marks a paired difference whose CI excludes zero. Each section states which radio scheme was used, so that NOMA stays visible in every result.

5.1 Physics checks (Gate G2)

An optimal altitude exists for every user class and moves lower as the rate target rises: at 0 dBm the best altitude is about 500 m for iot (10 kbps), 200 m for voice, 100 m for normal and 50 m for rescue and video users (fixed NOMA, dataset v2). Altitudes above 300 m are diagnostic only (the drone flies at 50-300 m).

Figure 5.1: Four training scenarios: crowds, scattered users, rescue users circled

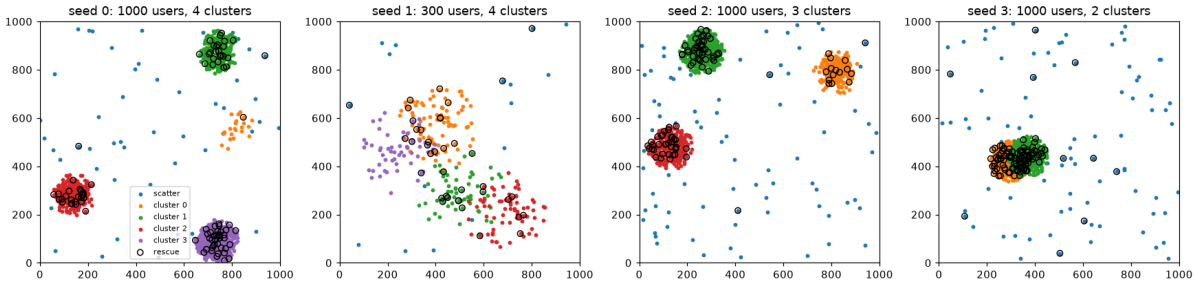


Figure: users_preview_v3.png

Figure 5.2: Coverage against altitude for every user class

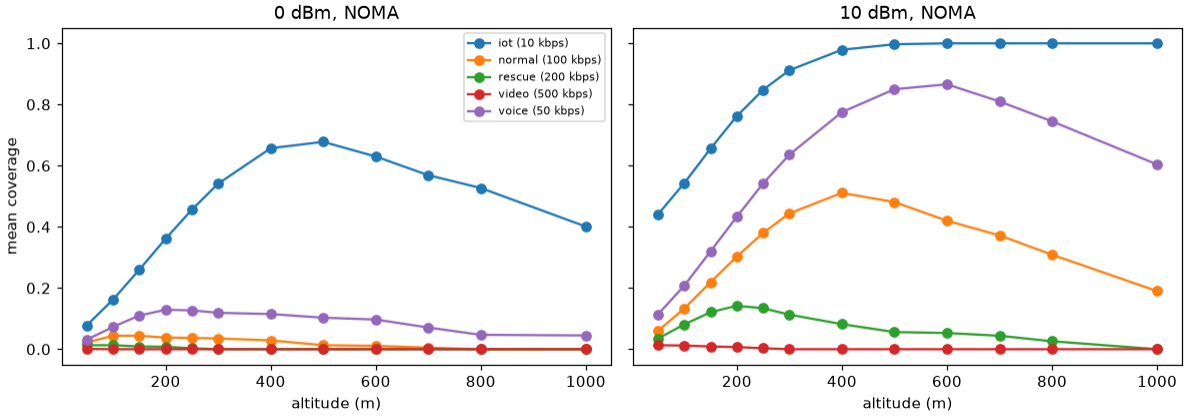


Figure: altitude_by_class_v2.png

5.2 H1: an adaptive drone beats a parked one

In the default 1 x 1 km scenarios with plenty of battery (16 Wh) a parked drone already covers 0.882 and the best pilot 0.914: little headroom. In the `hard` profile (2 x 2 km, 20 dBm, 3-8 buildings, sudden or drifting crowds, 16 Wh) the parked drone falls to 0.428 while the best pilots reach 0.740-0.741: **a 31-point gap, so H1 holds where the area is large and power scarce.** Neither low power alone nor a big map alone produces this gap; the combination does.

Table 5.1 H1: pilots on the `hard` *profile (100 scenarios, 16 Wh, all survive; fixed NOMA)*

Pilot	Coverage	Rescue users not served	Fairness
B0 fixed	0.428 +/- 0.042	0.846	0.504
B1 follow	0.627 +/- 0.027	0.535	0.538
B2 k-means	0.633 +/- 0.027	0.524	0.562
B3 grid	0.675 +/- 0.025	0.506	0.592
B4 PSO	0.740 +/- 0.023	0.417	0.709
B6 grid + altitude	0.741 +/- 0.023	0.419	0.715

Finding: most of B4's lead was altitude, not placement. B6 (grid search plus five altitudes) ties B4 (0.741 vs 0.740) and beats B3 by 0.066*.

Figure 5.3: Seven pilots on the hard profile

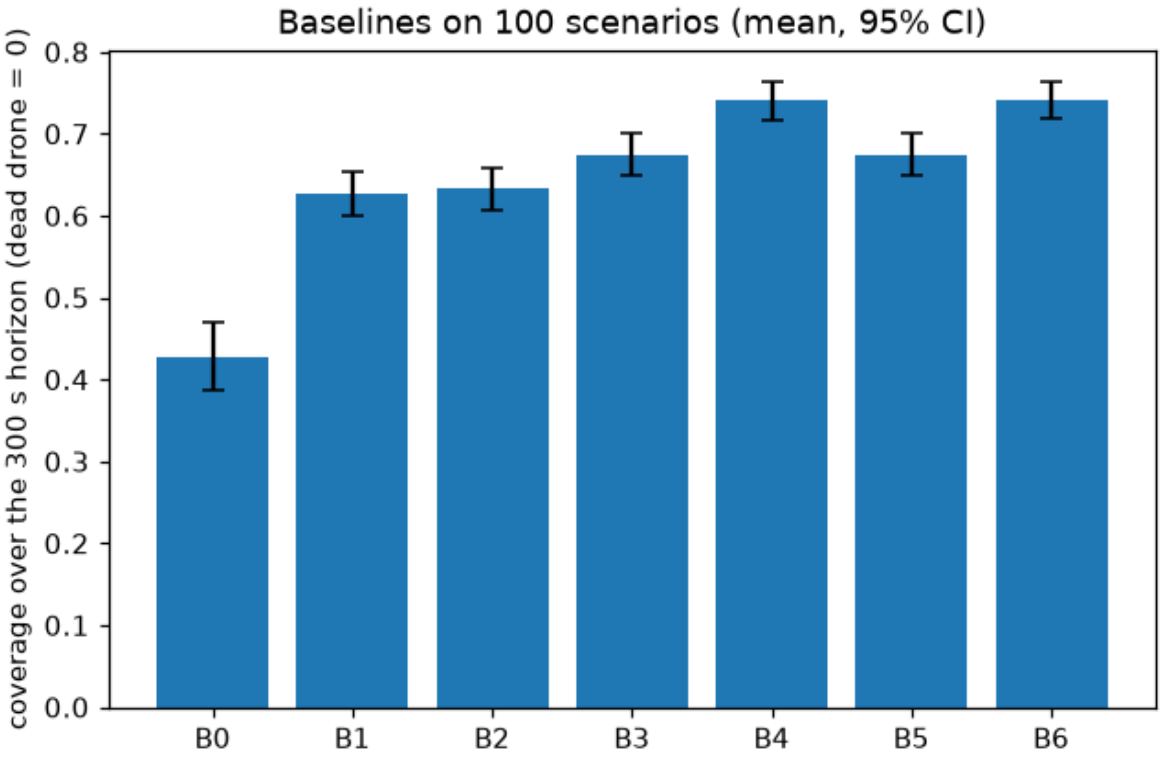


Figure: baselines_hard_7policies.png

Figure 5.4: Pilots on hard scenario 5 (600 users, 5 crowds, group drift): B0 0.49, B2 0.51, B3 0.55, B4 0.71, B6 0.71

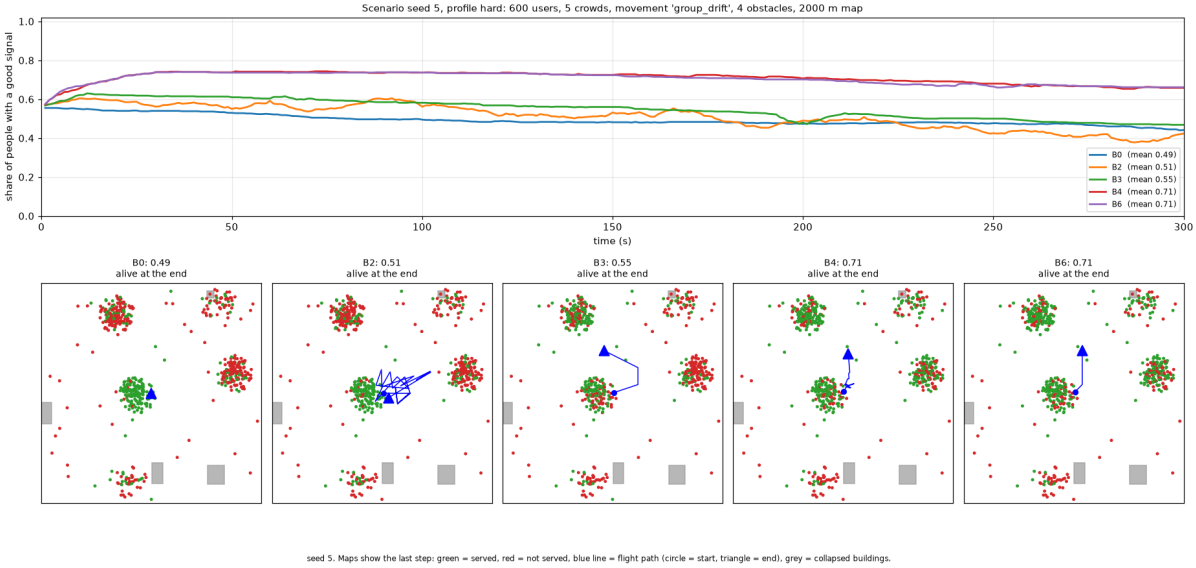


Figure: 1_hard_moving_crowds.png

5.3 H3: NOMA against OMA

Fixed NOMA (0.8/0.2 split) against OMA, inside the game. Every pilot played the same 100 scenarios under both schemes, with the search pilots evaluating spots under the scheme being tested.

Table 5.2 H3: fixed NOMA vs OMA in the game (paired NOMA minus OMA)

Profile	Pilot	Coverage NOMA / OMA / difference	Total rate (Mbps) NOMA / OMA
default, 30 dBm	B0	0.882 / 0.870 / +0.012*	275.1 / 228.8
default, 30 dBm	B3	0.909 / 0.929 / -0.020*	297.7 / 270.9
lowpower, 20 dBm	B0	0.797 / 0.745 / +0.052*	171.8 / 141.2
lowpower, 20 dBm	B3	0.865 / 0.808 / +0.058*	218.4 / 172.1
hard, 20 dBm	B2	0.633 / 0.568 / +0.065*	159.5 / 119.2
hard, 20 dBm	B3	0.675 / 0.616 / +0.059*	155.8 / 104.9

NOMA always gives more total rate (about 20-27% at 30 and 20 dBm) and serves 5-7 more people per 100 when power is scarce. At full power with a well-placed drone, OMA covers slightly more (1.7-2.0 fewer people per 100 with NOMA), and NOMA is less fair (Jain's index 0.10-0.20 lower).

Why fixed NOMA loses coverage at full power. With a fixed split the weak user always hears the strong user's signal, so SINR_weak < 0.8 / 0.2 = 4 however much power is used, and its rate is capped at (2 x 50 kHz) x log2(5) = **232 kbps**. A video user (500 kbps) in the weak half of a pair can never be served. This is the classic limitation of fixed power allocation and it motivated the adaptive scheme.

Figure 5.5: NOMA vs OMA in the game, hard profile

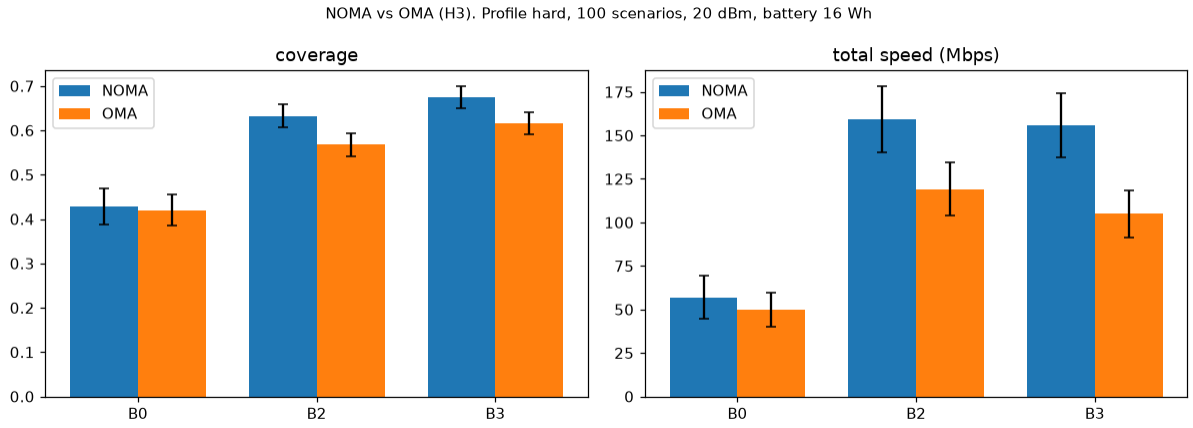


Figure: noma_vs_oma_in_game_hard_20dBm.png

Figure 5.6: NOMA pairs on the map and every user's rate (scenario 7, 1,000 users, 20 dBm): NOMA 0.81 vs OMA 0.76; the weak users sit on the 232 kbps ceiling

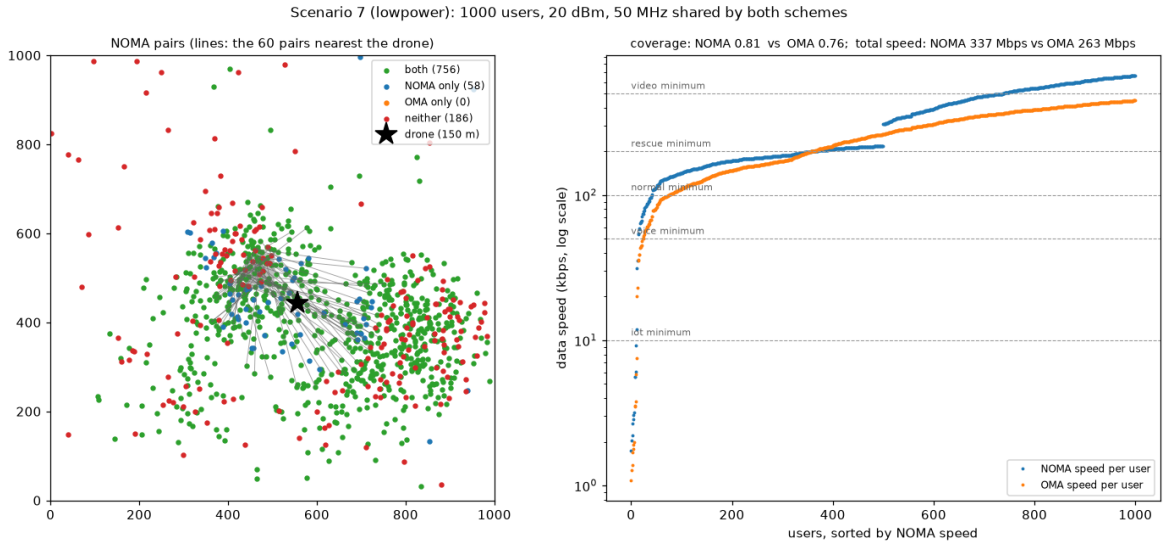


Figure: 7_noma_pairs_lowpower.png

Adaptive NOMA against equally optimised OMA. A fair test optimises both schemes the same way (Section 3.4).

*Table 5.3 H3: adaptive NOMA vs equally optimised OMA (100 scenarios, 16 Wh; * significant after Holm correction over 60 comparisons)*

Profile, pilot	OMA	NOMA fixed	OMA optimised	NOMA adaptive	Adaptive NOMA minus optimised OMA
lowpower, B0	0.745	0.797	0.893	0.944	+0.051*
lowpower, B2	0.769	0.835	0.924	0.962	+0.039*
lowpower, B3	0.808	0.865	0.945	0.983	+0.038*
hard, B0	0.420	0.428	0.658	0.690	+0.032*
hard, B2	0.568	0.633	0.777	0.811	+0.034*
hard, B3	0.616	0.675	0.808	0.849	+0.041*

- **Adaptive NOMA serves 3-5 more people per 100 than equally optimised OMA for every pilot**, with more total rate, better fairness and fewer unserved rescue users (B3 on lowpower: rescue outage 0.004 vs 0.030). The 232 kbps ceiling is gone.
- **Imperfect SIC** (eps = 0.05, hard): adaptive NOMA loses 0.014-0.024 coverage but still beats optimised OMA by +0.016 to +0.018*.

- **Rescue first** (hard): rescue outage falls sharply with adaptive NOMA (B3 0.203 to 0.117) for a small coverage loss (0.011-0.029); with optimised OMA the same rule costs 0.11-0.20 coverage.
- **The honest trade-off:** both optimised schemes serve far more people than the fixed ones (+0.12 to +0.26) but deliver about 55-65% less total rate and lower fairness than the fixed schemes. This is the price of the goal "serve as many people as possible", not of NOMA; the claim is "more people served, less total rate", not "better on every metric".

Figure 5.7: Four access schemes, hard profile

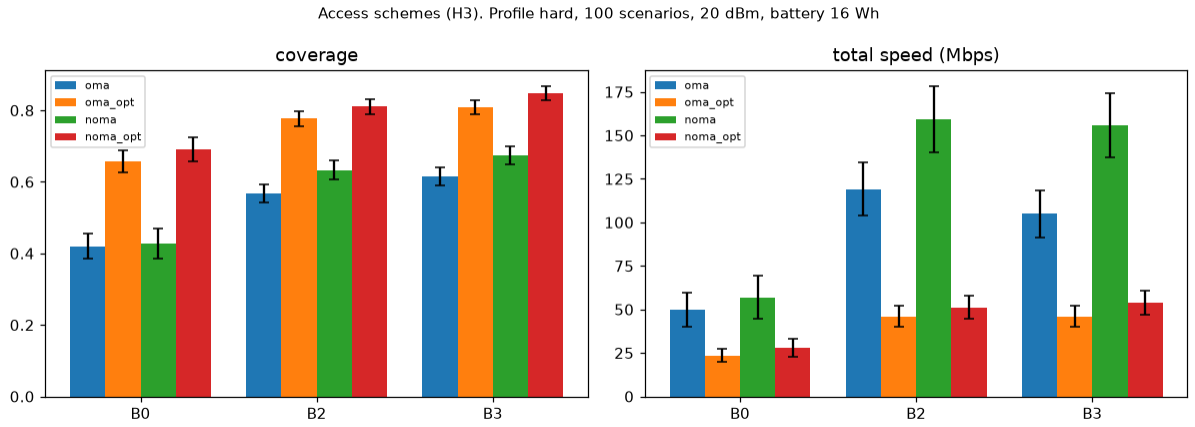


Figure: 8_access_schemes_hard.png

Figure 5.8: Four access schemes, low-power profile

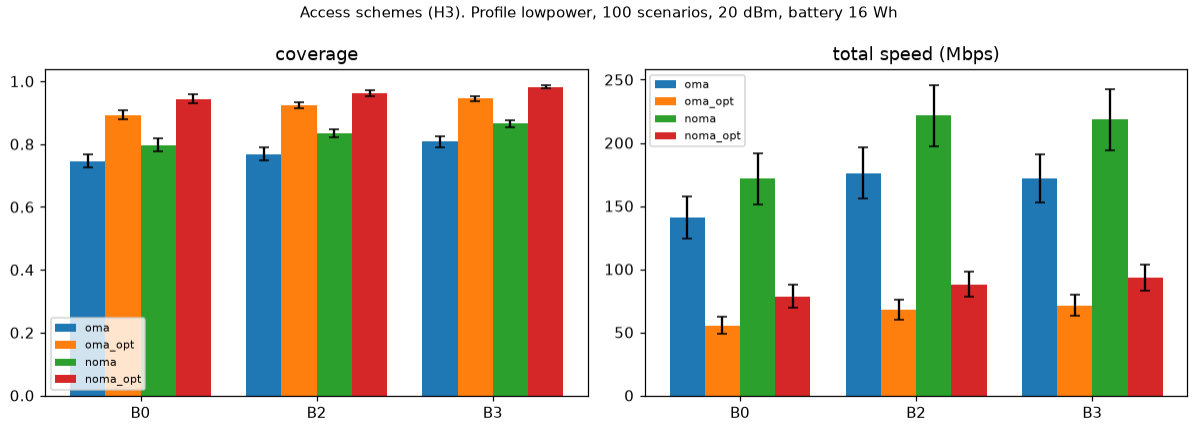


Figure: 9_access_schemes_lowpower.png

5.4 Energy: circling beats hovering on a tight battery

In the Zeng model the minimum-power speed is about 10.2 m/s (126.0 W against 168.5 W hovering). Pilot B7 chooses its spots like B6 but circles them at 9.8 m/s. With the acceleration limit and turning cost on, flying until the battery is empty (still crowds, 10 Wh, 50 scenarios):

Table 5.4 Circling vs hovering (fixed NOMA, full power)

Test	B6 (hovers)	B7 (circles, 50 m)	Difference
People-seconds served until the battery is empty	67,682	85,033	+25.6%*
Flight time	198 s	249 s	
Coverage while flying	0.697	0.696	same
5-minute mission, hard, 12 Wh	0.596 (never survives)	0.716	+0.120*
5-minute mission, hard, 13 Wh	0.652 (never survives)	0.730	+0.079*
5-minute mission, hard, 16 Wh	0.739	0.731	-0.009*

Figure 5.12: Circling vs hovering: 5-minute missions on a tight battery and flying until empty

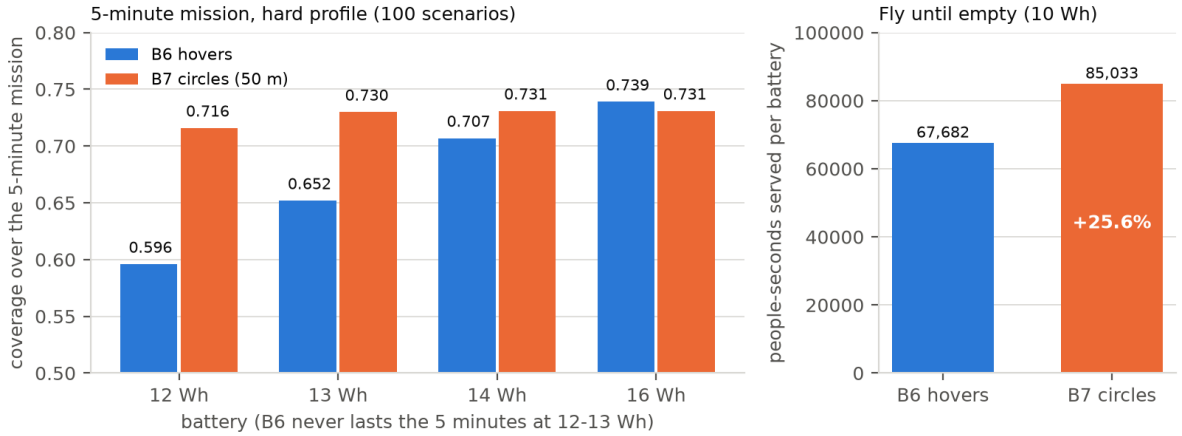


Figure: circling_vs_hovering.png

The gain stays between +21% and +31% when the unverified drag d0, the induced power, the acceleration limit or the radius are changed one at a time, so it does not depend on d0 = 0.6. A one-line rule (B8: circle once the battery would not last) captures the whole battery decision (12 Wh: 0.716; 13 Wh: 0.733), which means **a tight battery alone leaves nothing for an AI to win**. The turning-energy term is our own quasi-steady extension and is not validated against measurements.

5.5 Why the AI stalls on a still crowd: the information ladder

The search pilots score spots with every person's true position; the AI sees only coarse maps. To separate missing information from poor learning, three copies of B3's search were given less information.

Table 5.5 The information ladder (validation seeds 100-119, `rl_static`, fixed NOMA, paired with the AI E5a)

Pilot	What it knows	Coverage	vs the AI
B3	everything	0.659	+0.104*
B3obsUO	uncapped maps + buildings	0.629	+0.074*
B3obsU	uncapped maps	0.613	+0.059*
B3obs	exactly what the AI sees	0.605	+0.050*
B2 k-means	true positions	0.610	+0.055*
AI E5a	maps (learned)	0.555	
Random	nothing	0.430	-0.125*

About half of the AI's gap to B3 is missing information (0.054) and half is the AI not using what it sees (0.050). The AI is behind even on scenarios it trained on, so the problem is underfitting, not memorising. The key insight: B3 often parks **between** crowds so that one footprint reaches several of them; flying to the densest cell gives only 0.565, which is where every AI version stalled. The AI learned "go to the crowd", not "find the spot that reaches several crowds".

5.6 RL on a still crowd: Gate G4 (negative result)

Gate G4 asked the AI to beat random and match B2 on still crowds (`rl_static`, validation seeds, fixed NOMA). The pass mark was fixed before the final result: **the 3-run average must reach B2 - 0.03 = 0.580.**

Table 5.6 Gate G4 runs (validation, 20 scenarios)

Version	What changed	Coverage
First PPO	velocity control, decides every second	0.394 (below random 0.413)
`goto_hold10`	fly-to-spot control, decides every 10 s	0.525
E3	CNN map reader	0.549
E5a	CNN, 2.5x more training	0.555
E6 / E6b	exploration noise annealed	0.509 / 0.536
E0	uncapped maps, crowd size, own history, fresh scenarios, gamma 0.95 (3 runs)	**0.576 +/- 0.011**
E2	SAC	0.567
E3 (Track 2)	SAC + privileged critic (3 runs)	0.565 +/- 0.029
Bars	Random 0.430, B2 0.610, B3obs 0.605, B3 0.659	

Verdict: G4 not met (0.576 < 0.580). On a second unseen set (seeds 0-19) E0 averaged 0.519 against B2's 0.568 (-0.050*), so the gap is real. What worked: fly-to-spot controls and a 10-second rhythm (from below random to +0.095* above it), better inputs and fresh scenarios (0.555 to 0.576). What did not: a CNN, a finer map, longer training, annealed noise, SAC, and the privileged critic. The remaining fault is visible in Figure 5.11: the AI reaches the best spot at about 30 s and then flies on.

Figure 5.9: The first AI (velocity control) reaches the crowd and then flies off the map (seed 105: random 0.51, B0 0.53, B2 0.68, B3 0.78, PPO 0.40)

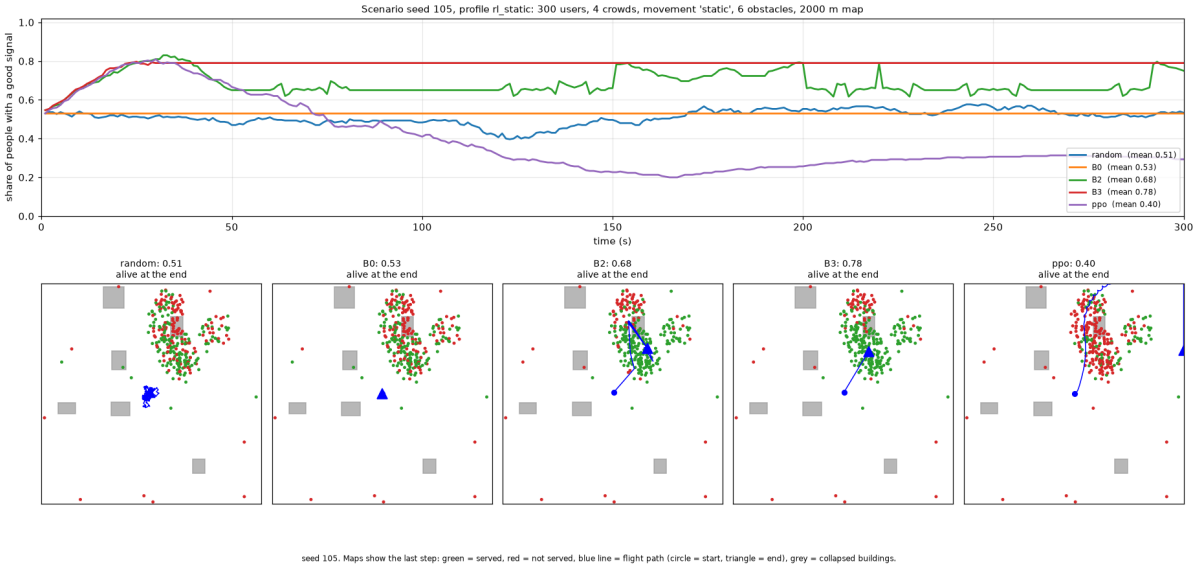


Figure: 2_still_crowd_AI_vs_pilots.png

Figure 5.10: Fly-to-spot control with a 10-second rhythm

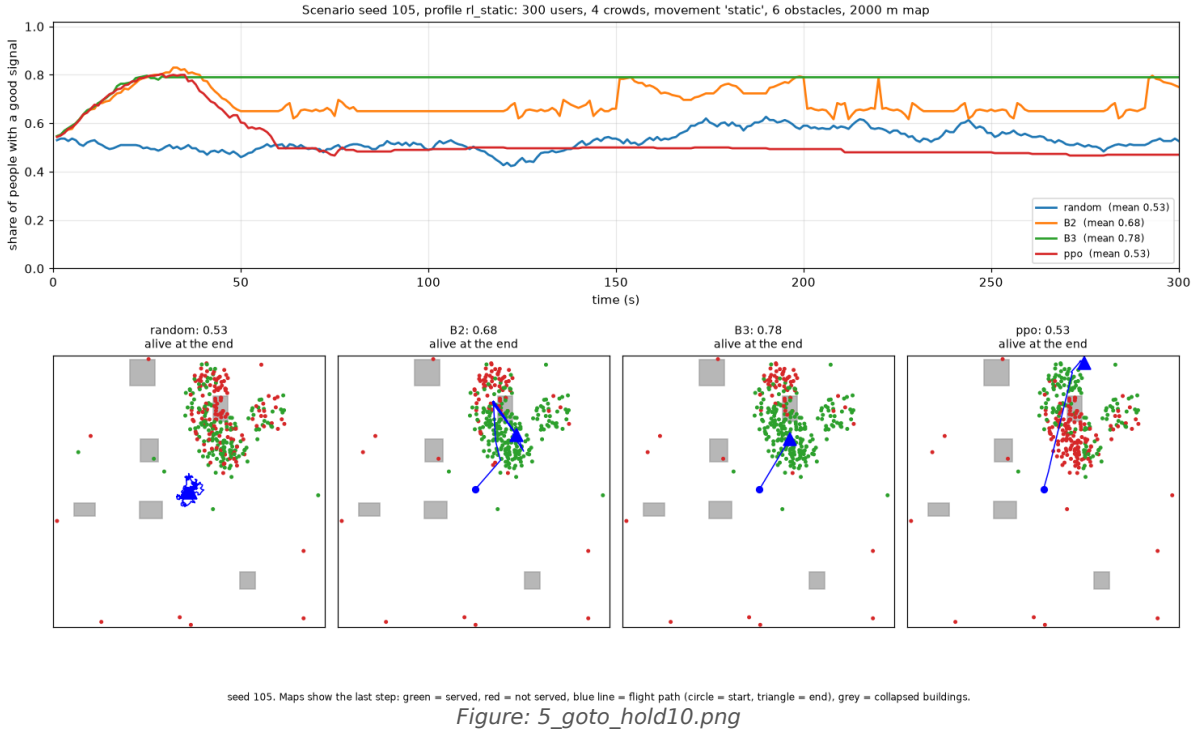


Figure: 5_goto_hold10.png

Figure 5.11: Best AI (E0) vs grid search B3 on scenario 105: the AI passes the best spot and settles at 0.66; B3 stays at 0.78

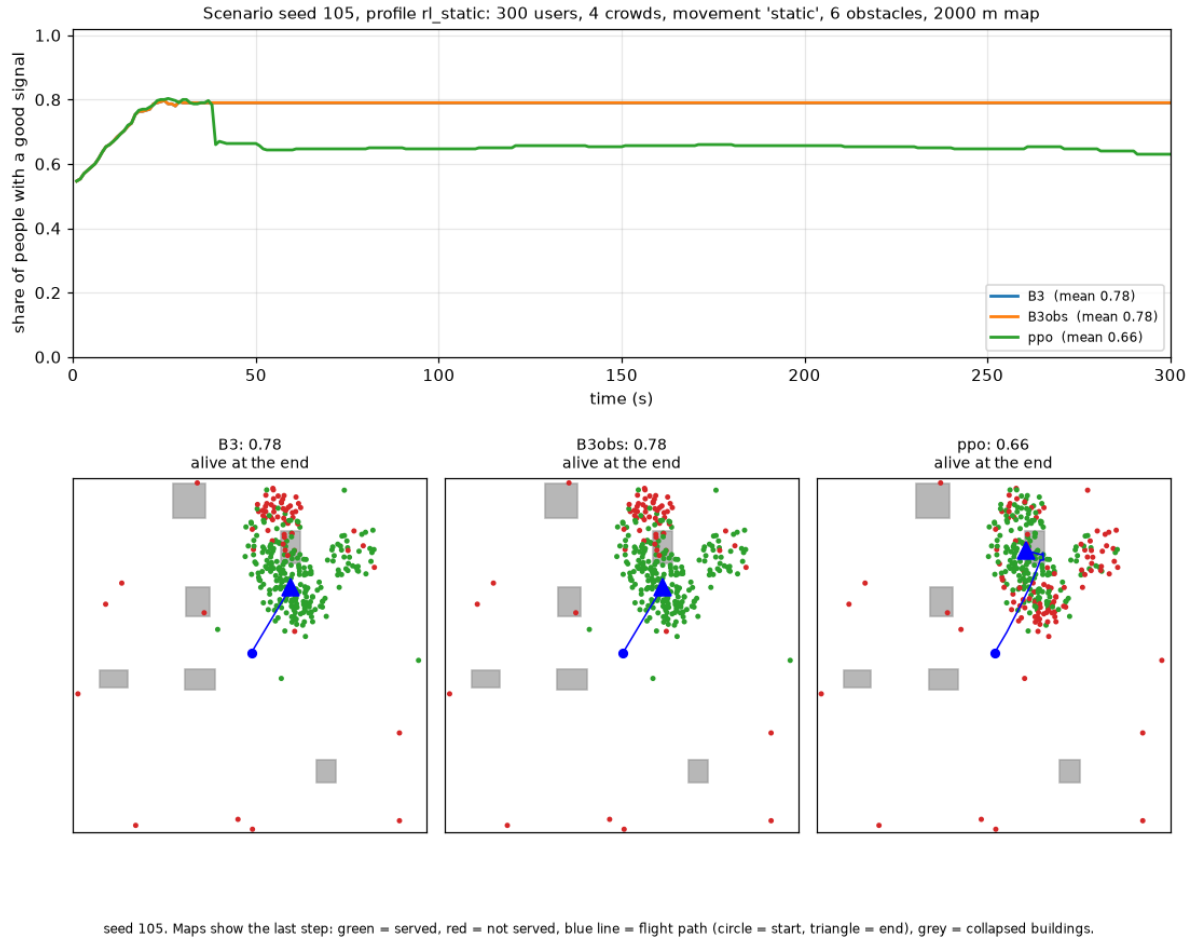


Figure: 10_e0_vs_b3_seed105.png

5.7 Is there room for an anticipating AI? Clairvoyant headroom and competing sites

H2 says an AI should win where it can anticipate. Before training on moving crowds we measured the prize: a clairvoyant pilot that is told the crowd's whole future plans the best possible flight by dynamic programming over B6's 980 spots, and its plan is flown in the real game. No AI can beat it, so its lead over B6 is the most anticipation could ever add. Pass mark fixed beforehand: at least 0.02 for some crowd type.

Table 5.7 Clairvoyant headroom on walking crowds (`hard`, 16 Wh, 120 scenarios, fixed NOMA, paired with B6)

Crowd type	B6	Clairvoyant	Gain
attraction (32)	0.776	0.778	+0.001
group drift (29)	0.745	0.747	+0.002
hotspot jump (26)	0.701	0.702	+0.001
surge (33)	0.748	0.750	+0.003
all (120)	0.745	0.746	+0.002*

Knowing the whole future is worth +0.002 (also +0.001 with adaptive NOMA): the drone (15 m/s, re-planning every 10 s) is much faster than walking crowds (0.5-3 m/s) and its footprint is hundreds of metres wide, so greedy re-planning keeps up. A pilot that looks only 30 s ahead even does worse than B6 (-0.011*). This saved weeks of training on scenarios where no AI could win.

Redesign: competing sites (pre-registered before any run). Survivors are at 2-3 sites 1.0-2.0 km apart, and half or more of the demand switches from site A to site B (a shelter opens, or a tower is repaired), at once or over 1 or 3 minutes. These runs use the G5 settings (adaptive NOMA, SIC residual 0.05, rescue first, acceleration limit).

Table 5.8 Room in the competing-sites designs (clairvoyant minus the best of B6, B7, B8; 12 scenarios per design; pass +0.02)

Sites apart (flight time)	Sudden switch	Over 1 minute	Over 3 minutes
1.0 km (67 s)	+0.025	+0.009	+0.005
1.5 km (100 s)	+0.072	+0.039	+0.020 (borderline)
2.0 km (133 s)	+0.108	+0.098	+0.056

7 of 9 designs passed on these design scenarios. **Room appears when the flight across is about as long as, or longer than, the switch.** As the pre-registered rules required, the designs were then re-measured on fresh scenarios, and a forecasting pilot was added to the bar: **B6fc** scores each candidate spot by the coverage it would give when the drone *arrives* there and for the following minute, using only what could be observed (positions now, movement in the last second and, in announced designs, the operator's message). Tests show it never reads the future.

Table 5.9 Room confirmed on fresh scenarios (seeds 1000-1019, 20 per design; first number: bar of B6, B7, B8; second: bar also including B6fc with the announcement; pass 0.02)

Sites apart	Sudden switch	Over 1 minute	Over 3 minutes
1.0 km	0.016 / 0.013 (no)	0.005 / 0.005 (no)	0.004 / 0.004 (no)
1.5 km	**0.056 / 0.038 (yes)**	**0.029 / 0.029 (yes)**	0.012 / 0.012 (no)
2.0 km	**0.091 / 0.046 (yes)**	**0.071 / 0.047 (yes)**	**0.029 / 0.029 (yes)**

Figure 5.13: Room for a pilot that thinks ahead, on fresh scenarios, against both bars

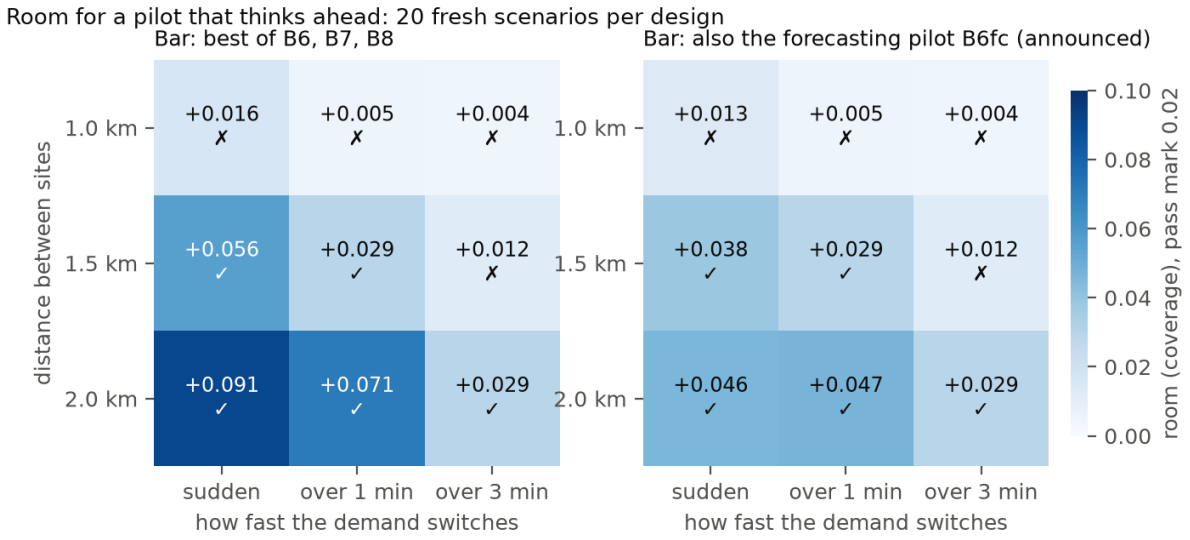


Figure: room_fresh_seeds.png

5 of 9 designs are confirmed; two that passed on the design scenarios (1.0 km sudden, 1.5 km over 3 minutes) did not, which is exactly why fresh scenarios were required. With a 13 Wh battery only the two 2 km designs (sudden, 1 minute) keep room. With the announcement, B6fc captures 36-52% of the room in the sudden and 1-minute designs, but in slow 3-minute switches it does worse than plain B6 because it leaves too early. A departure-time version (B6dep: stay until the best moment, then fly) narrowed the 1.5 km designs below 0.02 on 4 scenarios, leaving **the 2 km designs as the clear place where an AI could win (+0.04 to +0.09)**; these became the G5 training cells.

The anticipation rule (pre-registered; not confirmed). *Plain words.* Can we predict the room without running the clairvoyant pilot, from four numbers anyone can look up: the distance between the sites, the drone's top speed, how long the switch takes and how long the mission is? The idea: a drone that thinks ahead only gains when it must spend real time flying *between* the two places where people need it, covering neither, and when that dead time is a noticeable share of the mission.

Scientific statement. With footprint radius rho (where coverage halves, measured: 646-703 m), transit time T_tr = max(0, D - 2 rho) / v_max, A = (T_tr + 10 s) / (tau + 10 s) and M = T_tr / T, the rule selected by leave-one-out error on the 9 designs was **h = 0.0954 M^0.48 (A / (1 + A))^0.20** (fit R^2 = 0.986). The bar now also contains **B6dep**, a hand-written pilot that decides when to leave for the announced site; it is the best simple pilot in all 9 fitted designs, and with it in the bar the room at 1.5 km falls below 0.02 (0.010-0.015); only the 2 km designs keep room (0.026-0.036). Predictions for 8 new designs (other drone speeds, a 600 s mission, new distances and switch times; fresh seeds 2000-2011, 12 per design) were committed before they were run. Pass marks: R^2 >= 0.7 and at least 7 of 8 room / no-room calls right. All designs use adaptive NOMA (noma_opt, SIC residual 0.05, rescue first); no OMA run was made in this study, because it asks about timing, not access.

Table 5.10 The anticipation rule on 8 new designs (room h = clairvoyant minus the best simple pilot, 12 scenarios each; adaptive NOMA)

New design	Predicted h	Measured h (95% CI)	Call right?
top speed 8 m/s, 1.5 km, 60 s switch	0.017 (no room)	+0.031 +/- 0.018	no
top speed 25 m/s, 1.5 km, 60 s switch	0.009 (no room)	+0.007 +/- 0.005	yes
top speed 8 m/s, 2 km, sudden	0.049 (room)	+0.040 +/- 0.019	yes
top speed 25 m/s, 2 km, sudden	0.028 (room)	+0.019 +/- 0.008	no
600 s mission, 2 km, sudden	0.026 (room)	+0.018 +/- 0.006	no
600 s mission, 1.5 km, 60 s switch	0.009 (no room)	+0.010 +/- 0.005	yes
1.75 km, 30 s switch	0.024 (room)	+0.026 +/- 0.013	yes
1.25 km, 120 s switch	0.000 (no room)	+0.005 +/- 0.004	yes

Verdict: rule NOT confirmed. R^2 = 0.576 (pass 0.7) and 5 of 8 calls right (pass 7). The predictions follow the right trend (correlation 0.86), and the measured room still grows with the dead-transit share M, but the rule misses designs near the 0.02 line: it under-predicts the slow drone (8 m/s at 1.5 km) and over-predicts the fast drone and the long mission. A post-hoc check (not pre-registered) with a smaller footprint in the transit time got all 8 calls right but a far worse R^2 (below zero); it is exploratory and does not change the verdict. **What stands:** the clairvoyant test itself, which has to be run per design; the shortcut formula does not replace it.

Figure 5.14: The anticipation rule: room against the transit ratio A and the dead-transit share M, and predicted against measured room (blue: 9 fitted designs; orange: 8 new designs)

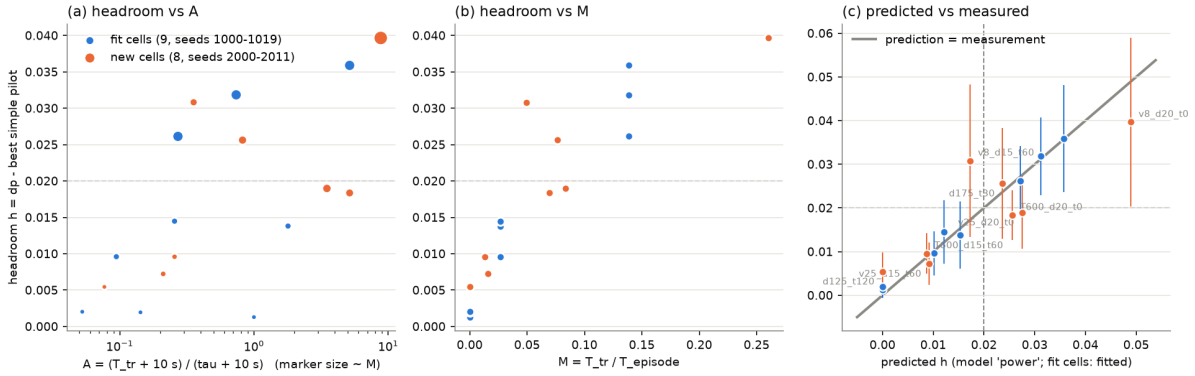


Figure: anticipation_rule.png

5.8 RL on competing sites: Gate G5, round 1 (negative result)

Round 1 (pre-registered, D-55) trained E0's design plus the event announcement, with fly-to-spot (`goto`) or move-by (`moveby`: "move this far", 0 = stay, built to stop the "flies past the spot" fault) steering, two training seeds each, 150,000 decisions per run (about 3.5 h each, on Kaggle), on the two 2 km announced cells. The bar is, per scenario, the best of B6, B7, B8 and B6fc. All runs use adaptive NOMA (eps 0.05, rescue first) and the acceleration limit; each agent was also evaluated with optimised OMA.

*Table 5.11 G5 round 1 (validation, 20 scenarios; * = CI excludes 0)*

Cell (bar)	Agent	Coverage (NOMA)	Agent minus bar	Coverage with OMA	NOMA minus OMA
sudden switch (0.756)	goto, seed 0	0.673	-0.083*	0.556	+0.117*
sudden switch	goto, seed 1	**0.680**	-0.076*	0.547	+0.133*
sudden switch	moveby, seed 0	0.671	-0.085*	0.588	+0.083*
sudden switch	moveby, seed 1	0.669	-0.087*	0.585	+0.084*
switch over 60 s (0.753)	goto, seed 0	0.673	-0.080*	0.555	+0.118*
switch over 60 s	goto, seed 1	**0.680**	-0.073*	0.552	+0.129*
switch over 60 s	moveby, seed 0	0.616	-0.137*	0.522	+0.094*
switch over 60 s	moveby, seed 1	0.629	-0.124*	0.528	+0.101*

Every agent is significantly below the bar (best 0.680 against 0.756 and 0.753), and also below plain B6 (by 0.03 to 0.10). The forecasting pilot B6fc (0.748 in both cells) carries almost all of the bar. The `moveby` steering did not help (equal on the sudden design, worse on the 1-minute design). **NOMA stays visible here too:** the same agents serve 0.08 to 0.13 more people with adaptive NOMA than with tuned OMA (all significant; the agents were trained with NOMA, so their OMA runs are outside what they learned).

What it means. The room exists, but an AI that has to learn *everything* (where to fly as well as when) from rough maps does not reach it with this budget; it does not even react as well as the search pilot. The room is mostly about **one decision: when to leave for the other site**. Round 2 of the full AI (5 more seeds) was therefore dropped as uninformative (D-56).

The hybrid timing AI (pre-registered; negative result). *Plain words.* Since the room is about one decision, we gave the AI only that decision: the search pilot B6 decides *where* the drone goes (position and altitude), and a small PPO policy decides every 10 s only "follow B6" or "leave now for the announced site". It is a hybrid and is reported as one (D-56).

Design. Same two 2 km announced cells, same bar as round 1 (per scenario, the best of B6, B7, B8, B6fc), 3 PPO training seeds per cell, 100,000 decisions each on 200 fresh training scenarios (Kaggle), the final model of every run evaluated on the 20 validation scenarios, no model selection. Pass mark: in at least one cell the 3-seed mean beats the bar with a Holm-corrected $p < 0.05$. A check confirmed that the hybrid that never leaves reproduces B6 exactly on all 40 scenarios. Adaptive NOMA (eps 0.05, rescue first), and every agent also flown with optimised OMA.

*Table 5.12 Hybrid timing AI (validation, 20 scenarios per cell; 3-seed mean; * = CI excludes 0)*

Cell	Bar	Clairvoyant	Hybrid (NOMA)	Hybrid minus bar	Hybrid minus B6	Hybrid with OMA	NOMA minus OMA
2 km, sudden switch	0.756	0.805	0.752	-0.004 +/- 0.013 (Holm p = 0.57)	+0.041 +/- 0.013*	0.620	+0.132 +/- 0.034*
2 km, switch over 60 s	0.753	0.798	0.734	-0.019 +/- 0.011* (Holm p = 0.001)	+0.014 +/- 0.007*	0.606	+0.128 +/- 0.031*

Verdict: not met (0 of 2 cells). The hybrid learned a real skill: it beats plain B6 by 0.041* and 0.014*, so it does leave early. But it only ties the bar on the sudden switch (the bar is mostly B6fc) and is significantly below it on the 60 s switch; it captures none of the clairvoyant room (-8% and -44%). Because Round A did not pass, the two extra seeds were not run, as pre-registered. For comparison, the hand-written departure pilot B6dep, built for the anticipation study after the hybrid's bar was fixed, scores 0.780 and 0.774 on the same validation scenarios, above both the bar and the hybrid. NOMA stays visible: the same hybrid serves 0.13 more people per scenario with adaptive NOMA than with tuned OMA.

Where the full AI's shortfall comes from. The anticipation study split the gap between the clairvoyant pilot and the best round-1 agent (goto, seed 1) into five parts, each a paired difference between two pilots on the same 20 validation scenarios.

Table 5.13 The AI's shortfall split into parts (G5 cells, validation; paired mean +/- 95% CI; adaptive NOMA)

Part	Pilots compared	2 km, sudden	2 km, over 60 s
Foresight	clairvoyant (0.805 / 0.798) minus B6dep (0.780 / 0.774)	+0.024 +/- 0.008	+0.023 +/- 0.008
Using the announcement	B6dep minus B6 (0.711 / 0.721)	+0.069 +/- 0.016	+0.054 +/- 0.013
Choosing the altitude	B6 minus B3 (the AI's fixed 150 m)	+0.019 +/- 0.012	+0.025 +/- 0.010
Information	B3 minus the same search on the AI's maps	+0.028 +/- 0.018	+0.028 +/- 0.016
Learning	that search minus the best agent (0.680)	-0.016 +/- 0.022	-0.012 +/- 0.019
Total	clairvoyant minus the best agent	**+0.125 +/- 0.019**	**+0.118 +/- 0.015**

In plain words: the AI is **not** a poor learner here; it matches a search that sees what it sees (the learning part is about zero and not significant). It loses because it was not given the altitude choice, sees only rough maps, and above all does not use the operator's announcement the way a hand-written departure rule does (the largest part, 0.054-0.069).

5.9 When the textbook channel is wrong: model mismatch (negative result)

Plain words. The search pilots choose spots by scoring them with the same radio model the game uses, which is an unfair advantage over any real drone. Here the game's TRUE radio world differs from the textbook model the pilots plan with: signal shadowing that is correlated over tens of metres, a 25 dB building loss where the pilots assume 15 dB, and buildings missing from the pilots' map. An AI that learns only from the coverage it measures might then beat pilots that trust the textbook.

Design (pre-registered). Still crowds (`rl_static`) with the G5 settings. Room = B6true (the search on the true channel) minus the best of B6 and B6meas (a pilot that picks a few spots with the textbook model, then flies there and measures); a design has room if the room is at least 0.02 and its CI excludes 0. The AI (E0's design, with altitude control `gotoz`, 3 training seeds x 150,000 decisions, Kaggle) must beat the bar (best of B6, B6meas, B3 per scenario) with a Holm-corrected $p < 0.05$; on the no-mismatch control it must not beat B6.

Table 5.14 Model mismatch: room test (validation, 20 scenarios; adaptive NOMA)

Design	B6	B6meas	B6true	Room [95% CI]
no mismatch (control)	0.865	0.857	0.869	+0.004 [+0.002, +0.006]
shadowing 4 or 8 dB, correlation 25-150 m (6 designs)	0.838-0.862	0.836-0.858	0.855-0.876	+0.008 to +0.016 (none passes)
building loss 25 dB	0.836	0.835	0.852	+0.010 [+0.005, +0.015]
2 / 4 hidden buildings	0.846 / 0.817	0.844 / 0.829	0.860 / 0.853	+0.010 / +0.017 (none passes)
8 hidden buildings (exploratory)	0.752	0.786	0.829	+0.036 [+0.023, +0.050]
all errors together (exploratory)	0.765	0.790	0.832	+0.035 [+0.020, +0.050]

The pre-registered grid had no room (largest +0.017). Two larger doses were added *after* seeing that null and are labelled exploratory everywhere; their room was confirmed on fresh seeds 1000-1019 (+0.026 and +0.033) before any training.

Table 5.15 Model mismatch: the AI against the bar (validation, 20 scenarios, 3-seed mean; 95% CI in brackets)

Design	Bar	AI (NOMA)	AI minus bar	AI minus B6	NOMA minus OMA (AI)
8 hidden buildings (exploratory)	0.795	0.756	-0.039 [-0.061, -0.016]	+0.004 [-0.036, +0.043]	+0.111 [+0.078, +0.143]
all errors together (exploratory)	0.801	0.700	-0.101 [-0.138, -0.064]	-0.066 [-0.101, -0.030]	+0.078 [+0.053, +0.103]
no mismatch (control)	0.865	0.817	-0.048 [-0.066, -0.031]	-0.048 [-0.066, -0.030]	+0.076 [+0.051, +0.100]

Verdict: the AI does not beat the bar in any design, every training seed loses, and a fresh draw of the hidden errors gives the same picture (-0.046 and -0.090), so it is not memorising. The control behaves as required. The one good sign: where the textbook is wrong (8 hidden buildings), plain B6 loses about 0.11 and the AI ties B6, so the AI is not fooled by the wrong model; it is still beaten by the pilot that **measures and searches** (B6meas), which recovers most of the loss (0.752 to 0.786, ceiling 0.829). This was the outcome written down before training (the room, 0.026-0.036, is smaller than the AI's known learning gap of about 0.08). The useful lesson is model-based, not an AI result: **when the radio model may be wrong, measure a few spots and then search.** NOMA stays visible: the same agents serve 0.08 to 0.11 more people with

adaptive NOMA than with tuned OMA.

5.10 Sim-to-sim: does the simplified flight model change the results?

The same pilots flew the same validation scenarios (seeds 100-104) in the point-mass game and with the simulated quadcopter and autopilot (Section 3.10), in calm air and in 5 m/s wind with gusts.

Table 5.16 Sim-to-sim check (5 scenarios per cell)

Cell	Game coverage	Quadcopter, calm	Quadcopter, wind	Coverage gap (calm / wind)
B6 on `hard`	0.694	0.691	0.691	-0.003 / -0.003
B6 on `sites_d20_t0_ann`	0.632	0.626	0.626	-0.006 / -0.006
AI E0 on `rl_static`	0.447	0.436	0.435	-0.010 / -0.012

Flying with a quadcopter model, autopilot lag and a 30 degree tilt limit costs only 0.003 to 0.012 coverage, and wind changes the result by at most 0.002 (the autopilot holds its spot within about a metre). Pilots built for the point game would overshoot their target on a realistic drone unless they brake in time; with the acceleration limit they do (median 1.8 m off target). So **the coverage conclusions carry over to a flying vehicle model** (a simulation-to-simulation check, not a real flight). **The quadcopter's energy numbers are not trustworthy yet:** the Zeng formula makes wobbling and wind look cheaper than hovering, so battery results on the quadcopter need more work. Only 5 scenarios per cell, and not validated against a real drone or ArduPilot/PX4 software-in-the-loop.

Figure 5.15: The 3D flight page: B6 flown by the simulated quadcopter in wind on a 2 km competing-sites scenario (validation seed 100), 170 s into the mission, crossing towards the newly active site (grey = people not counted now, at the other site); one scenario, for illustration only

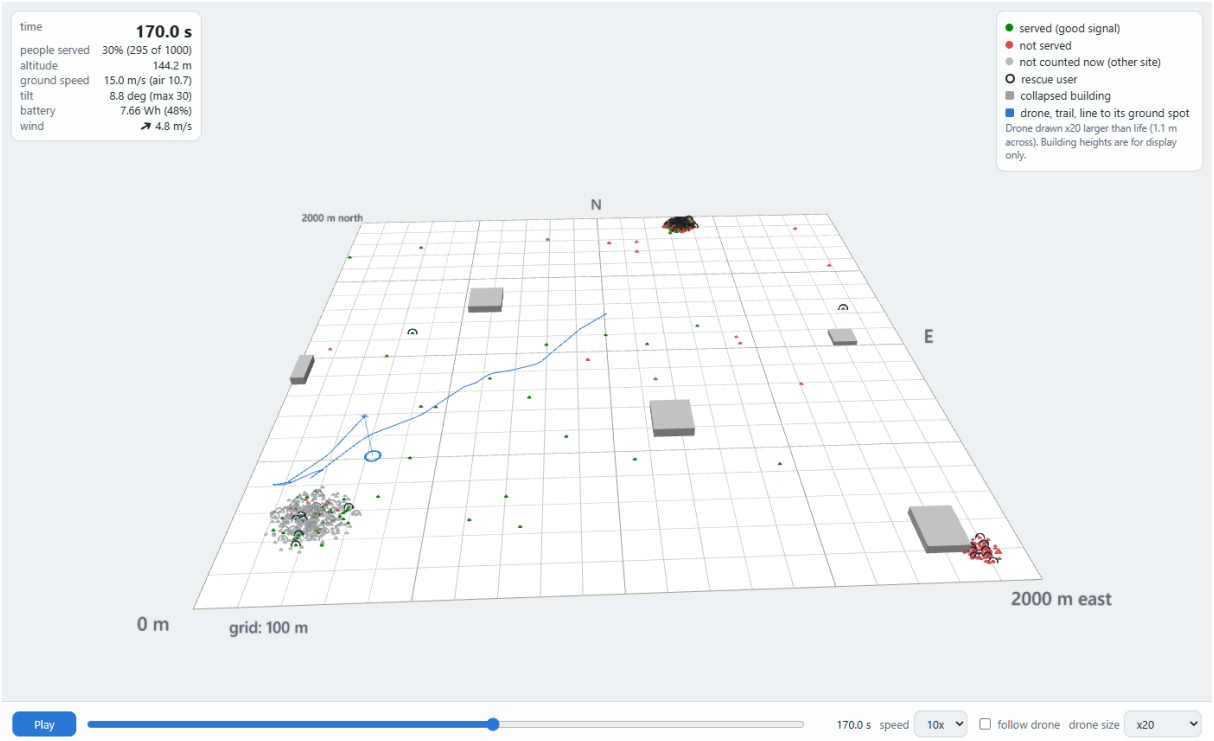


Figure: fig_3d_flight.png

5.11 Value of location information (pre-registered; passed)

Plain words. Every search pilot so far knew where every person stands, which no real drone does. But after a disaster the drone *is* the network, so a phone that attaches to it can send its GPS

position. In practice only some phones do, GPS has an error, and reports arrive every so often. **What share of phones must report, how accurately and how often, for the planner to keep at least 98% of the coverage it gets with full information?**

Design (pre-registered before any main run). B6 (and B3 as a secondary planner) searched on a *belief* built from the reports: a share f of the phones reports (0.05 to 1), each report has a GPS error of sigma = 0, 10 or 50 m per axis, and phones report every 0, 30 or 60 s (unsynchronised). The k visible reporters stand for k / f people. The drone then serves the TRUE people. 30 scenarios per configuration (validation seeds 100-119 and fresh seeds 1000-1009), the same for every configuration, 1,440 episodes, all surviving the 300 s. Ratio R = coverage / full-information coverage, paired bootstrap CI. f* = the smallest share whose R, and that of every larger share, is at least 0.98. **Pass:** for B6, on both profiles and at both 10 m and 50 m error, f* is identifiable and its 95% interval stays below 1. Radio: fixed NOMA on `hard`, adaptive NOMA on the competing-sites cell (`sites_d20_t60_ann`, G5 settings); no OMA run was made, because the study is about information, not access.

Table 5.17 Value of location information (B6; ratio to full information [95% CI]; reports every second)

Share of phones reporting (about how many)	`hard`**, 10 m error**	`hard`**, 50 m error**	**competing sites, 10 m error**	**competing sites, 50 m error**
0.05 (20)	0.957 [0.938, 0.970]	0.958 [0.941, 0.972]	0.983 [0.963, 1.004]	0.987 [0.964, 1.008]
0.1 (39)	0.973 [0.961, 0.982]	0.972 [0.961, 0.981]	0.986 [0.965, 1.006]	0.990 [0.970, 1.008]
0.2 (79)	0.978 [0.968, 0.985]	0.979 [0.971, 0.986]	1.004 [0.992, 1.018]	1.003 [0.990, 1.017]
0.5 (197)	0.990 [0.986, 0.994]	0.988 [0.984, 0.992]	0.998 [0.987, 1.009]	1.001 [0.990, 1.012]
1 (393)	1.000 [0.999, 1.001]	0.994 [0.991, 0.997]	1.007 [1.002, 1.011]	1.007 [1.000, 1.015]
f* (95% interval)	**0.5 [0.1, 0.5]**	**0.5 [0.1, 0.5]**	**0.05 [0.05, 0.2]**	**0.05 [0.05, 0.2]**
Full-information coverage	0.770	0.770	0.717	0.717

Verdict: PASS. What it means:

- **GPS accuracy hardly matters** at this scale: 50 m error costs nothing measurable (B6, 20% reporting: 0.978 at 10 m vs 0.979 at 50 m). The planner needs to know roughly *where the crowds are and how big*, not where each person stands; its search grid is 154 m wide.
- **The share of phones matters on moving crowds** (`hard`: 5% reporting keeps 95.7%, 20% keeps 97.8%, 50% keeps 99.0%), much less on competing sites, where every crowd is a dense site that even 5% of phones (about 20 reports) reveal (98.3%).
- **Report age matters when people move away in an event:** on competing sites at 20% reporting, B6 drops from 100.4% (every second) to 98.4% (every 30 s) and 97.8% (every 60 s); B3 from 98.7% to 96.3%. On `hard` the effect is small (97.8% to 97.5%).
- Ratios slightly above 1 (up to 1.007) occur because B6 is a greedy planner, not an optimal one: a perturbed picture can by chance lead it to a slightly better spot. It is not "better information".

Design rule for a rescue team (within this simulator): get position reports from about half of the attached phones, refreshed at least every 30 s; GPS quality down to about 50 m is enough. With one phone in five reporting expect about 2% less coverage on moving crowds, and with one in twenty about 4% less. The assumptions behind it are named, not verified: GPS errors independent per report, reporting independent of position and class (in reality older phones, dead batteries and people trapped indoors report less), and the share f known to the planner.

Figure 5.16: How many phones must report? Coverage relative to full information against the share of phones that report, on `hard` and on competing sites (30 scenarios, 95% CI)

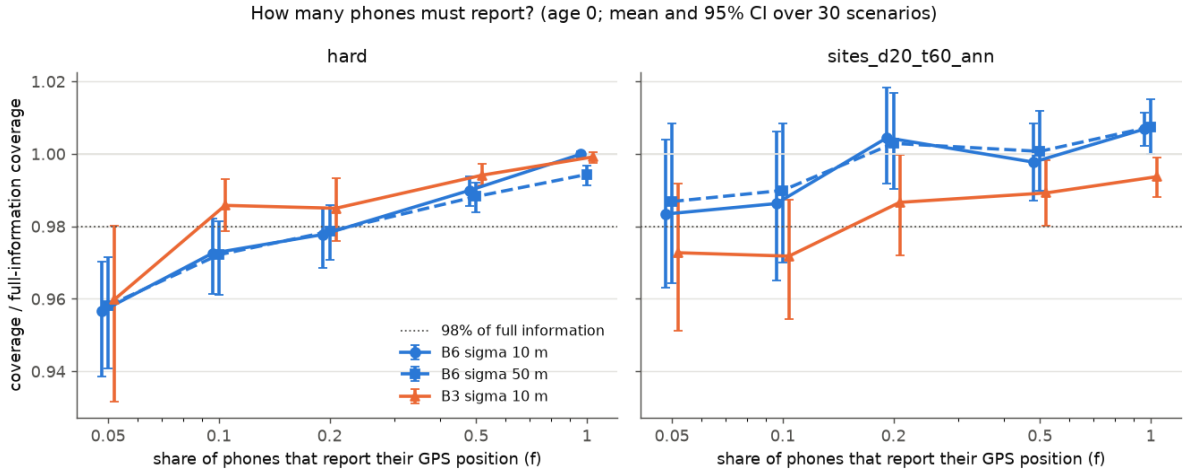


Figure: voi_ratio_vs_f.png

5.12 A scenario based on the September 2024 Kathmandu floods

Plain words. We took a real 2 x 2 km piece of Kathmandu along the Bagmati river (Teku and Kuleshwor, with Balkhu at the west edge and Thapathali at the east edge), downloaded every building footprint from OpenStreetMap [38] (12,061 footprints; map data (c) OpenStreetMap contributors, ODbL), gave each building a height, and let the same pilots serve a flood crowd there. **The buildings are real; the people are reconstructed** from the map with engineering assumptions (stranded on roofs within an assumed 200 m flood band, at shelters, on dry ground, scattered; an evacuation from roofs to shelters). It is a scenario *based on* the floods, not real data and not a digital twin. Riverside flooding in Balkhu and at Thapathali in late September 2024 is taken from news reports [39], [40]; Teku and Kuleshwor themselves were not verified as flooded from a primary source, and the flood band is a fixed-width strip, not the 2024 flood extent.

Design. 20 crowd draws (seeds 0-19), pilots B0, B2, B3, B6, B7, B8, B6fc, each with adaptive NOMA and with equally optimised OMA (equal bandwidth and power); the same pilots and settings on the synthetic `hard` map (same seeds) for comparison. G5 settings in both (SIC residual 0.05, rescue first, acceleration limit, 20 dBm, 16 Wh). The height-aware line of sight (Section 3.3) replaces the statistical one.

*Table 5.18 Kathmandu case: pilots, NOMA and OMA (20 scenarios, mean +/- 95% CI; paired NOMA minus OMA; * = CI excludes 0)*

Pilot	Kathmandu NOMA	Kathmandu OMA	Kathmandu NOMA minus OMA	`hard` NOMA	`hard` OMA	`hard` NOMA minus OMA
B0 parked	0.674 +/- 0.032	0.479	+0.195 +/- 0.012*	0.560 +/- 0.093	0.369	+0.191 +/- 0.057*
B2 k-means	0.713 +/- 0.033	0.521	+0.191 +/- 0.009*	0.726 +/- 0.054	0.568	+0.159 +/- 0.037*
B3 grid	0.741 +/- 0.032	0.577	+0.164 +/- 0.010*	0.769 +/- 0.054	0.633	+0.135 +/- 0.038*
B6 grid + altitude	0.781 +/- 0.025	0.654	+0.127 +/- 0.008*	0.837 +/- 0.044	0.773	+0.064 +/- 0.021*
B7 circling	0.774 +/- 0.025	0.626	+0.148 +/- 0.007*	0.833 +/- 0.045	0.761	+0.072 +/- 0.021*
B8 = B6fc (= B6 here)	0.781 +/- 0.025	0.654	+0.127 +/- 0.008*	0.837 / 0.836	0.773 / 0.777	+0.064* / +0.059*

What the numbers say:

- **The ranking of the pilots does not change on the real map** (B6 = B8 = B6fc > B7 > B3 > B2 > B0, with NOMA and with OMA), exactly as on the synthetic map. Every gap to B6 is significant (B0 -0.107*, B2 -0.069*, B3 -0.041*, B7 -0.008*).
- **H1 is weaker here** (B6 minus parked B0: +0.107 against +0.278 on `hard`): the river runs close to the map centre and nobody walks away, so a parked drone at the centre already sees most people.
- **H3 is stronger here: NOMA helps about twice as much on the real city for the searching pilots** (B6 +0.127 against +0.064, difference +0.063 +/- 0.022*; B7 +0.148 against +0.072). For B0, B2 and B3 the gains are similar (difference not significant). Our reading, not tested separately: the height-aware line of sight splits the people into clear and blocked links whose channels differ by about 19 dB, and NOMA gains most when paired users have very different channels.
- **Who gains:** with NOMA, B6 serves 0.83 of the people stranded on roofs and 0.68-0.77 of those at shelters, on dry ground and scattered; with OMA the ground-level groups fall to 0.52-0.60. Rescue users are served at 0.89 with both schemes (rescue first).
- **Circling** (B7) costs 0.008 coverage and saves 2.2 Wh (3.64 vs 1.41 Wh left), +17.6% people-seconds per Wh. Forecasting adds nothing (the evacuation is unannounced and people stand still), so this case could not serve as a test of the anticipation rule.
- **The line-of-sight model is the biggest assumption:** keeping Al-Hourani's statistical LoS and adding 15 dB on blocked rays (which counts the buildings twice) lowers B6 from 0.781 to 0.636, but the NOMA gain stays (+0.13 to +0.14). The flood-band width (150 or 300 m) moves B6 by less than 0.02.
- **Altitude and the law:** B6 flies at about 290 m, near the top of the 50-300 m range, because from high up the rays clear the roofs. **That is above the 120 m limit commonly set for routine drone flights** [VERIFY: India's Drone Rules 2021; Nepal's rules not checked], so this result needs a re-run with a 120 m ceiling before it could inform a real plan.
- Sim-to-sim on the real map (one scenario, seed 0): the simulated quadcopter flying B6 scores 0.8145 against 0.816 in the game.

Figure 5.17: Kathmandu, Teku / Kuleshwor reach of the Bagmati: buildings shaded by height, the river, the assumed 200 m flood band, shelter candidates and the reconstructed people of seed 0 (a scenario based on the Sept 2024 floods; not real data; map data (c) OpenStreetMap contributors)

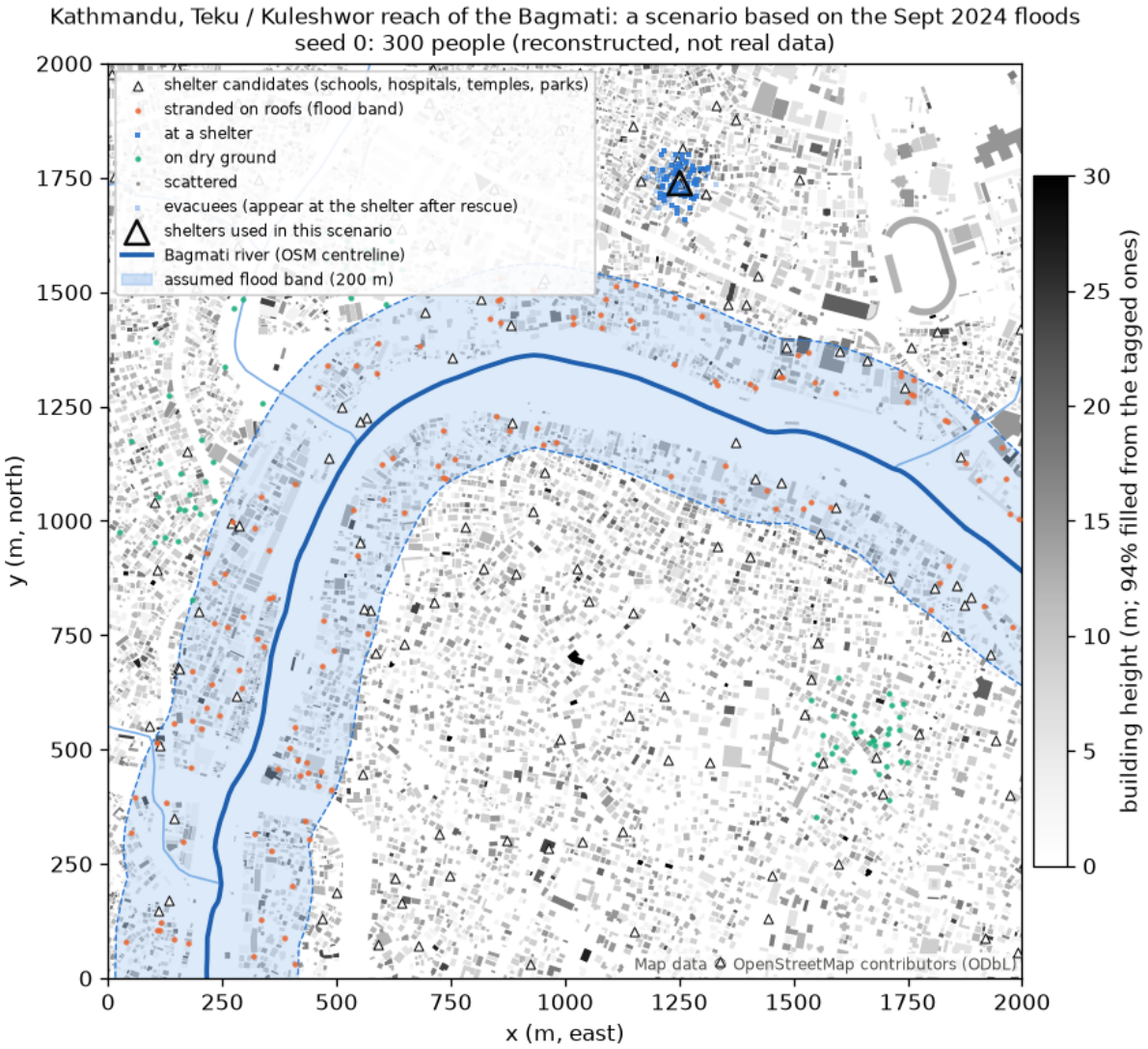


Figure: nepal_map.png

Figure 5.18: NOMA vs OMA at equal bandwidth and power: Kathmandu place profile and synthetic `hard` with the same settings (20 scenarios, 95% CI; map data (c) OpenStreetMap contributors)

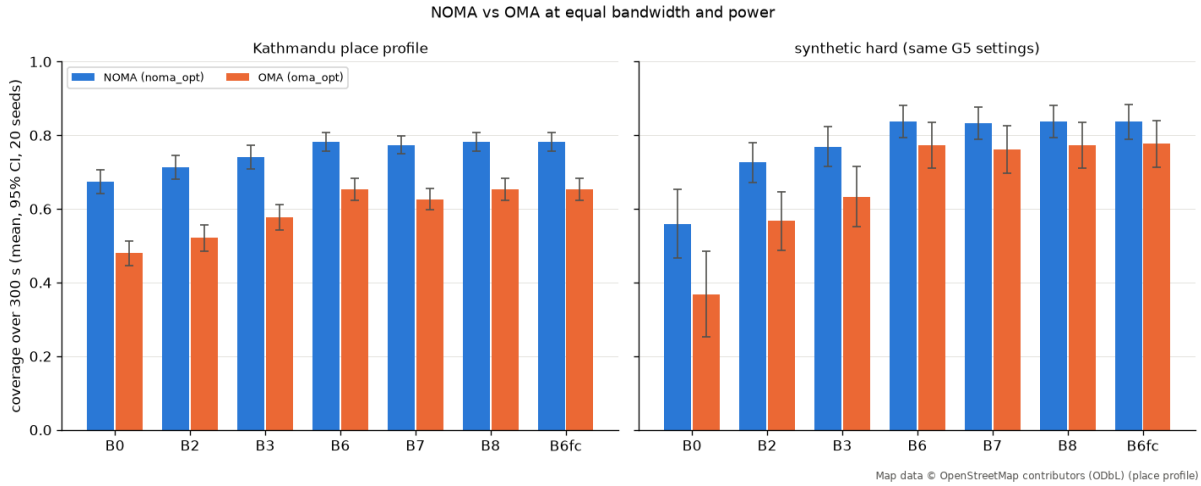


Figure: nepal_bars.png

Limits of this case: 94% of the building heights are drawn from the 6% that are tagged; no diffraction, trees, terrain or indoor users; only 20 crowd draws; the comparison of NOMA gains between the two maps is unpaired (different scenarios); the geometric line of sight has not been checked against a measurement in Kathmandu.

Chapter 6. Discussion

6.1 Where learning helps and where it does not

The results support a split of the control problem by time scale, which a communications engineer would make anyway.

Table 6.1 Where each tool fits

Layer	Time scale	Best tool in our results
Radio resource allocation (pairing, power per pair)	every second	Closed form and optimisation: adaptive NOMA (Section 5.3)
Placement for the users as they are now	every 10 s	Search: B6 (grid + altitude) is near-optimal for walking crowds (Section 5.7)
Battery strategy (hover or circle)	minutes	A one-line rule: B8 (Section 5.4)
Mission timing when demand jumps between distant sites	minutes	The only place with confirmed room (2 km designs); a hand-written departure rule (B6dep) takes most of it, leaving +0.023 to +0.024 for perfect foresight; neither the full AI nor the hybrid timing AI reached the simpler forecasting bar (Section 5.8)
Placement when the radio model is wrong	every 10 s	Measure a few spots, then search (B6meas); the AI did not beat it (Section 5.9)
Where people are	minutes	Phone position reports: about half of the phones, refreshed every 30 s, 50 m GPS error is enough (Section 5.11)

"Do not learn what you can compute." On still crowds, a search with the AI's own inputs reaches B2's level, so a learner could in principle pass G4, but end-to-end RL did not turn the maps into a precise spot. On walking crowds no controller can gain more than 0.002 over greedy re-planning. Where room exists (competing sites, 2 km), a forecasting pilot already captures most of it, a departure-time rule captures more, and neither the full AI nor the hybrid did as well. Where the textbook channel is wrong, a pilot that measures and searches beat the AI that learns from measurements. This is the "where it does not help" half of H2, which the hypothesis explicitly asks us to report.

6.2 H2: the final verdict

H2 is **not supported** by our evidence. Four pre-registered tests of learning, four negative results:

Test	Room for any pilot	Best AI against the bar
G4, still crowds	not measured by a clairvoyant; B2 0.610, B3 0.659	0.576 vs pass mark 0.580: not met
G5 round 1, full AI, 2 km competing sites	clairvoyant +0.048* / +0.044* over the bar	-0.076* / -0.073*
Hybrid timing AI, same cells and bar	the same	-0.004 (n.s.) / -0.019*
Model mismatch (exploratory cells)	+0.036* / +0.035*	-0.039* / -0.101*

The clairvoyant method explains part of this (no room on walking crowds) and located the one regime where room exists; the shortfall decomposition (Table 5.13) shows that the AI's learning is not the main problem there, its inputs and its use of the announcement are. The anticipation rule that was meant to predict the room without the clairvoyant test was **not confirmed** (5 of 8 calls right). The honest framing is therefore that **physics-based search is the main controller; learning, with our inputs and budgets, did not help anywhere we tested; and we can measure in advance where it could**.

Plain words. We asked an AI to learn four different jobs, each time with a pass mark written down first. It never beat the best hand-written pilot. We report this as a finding, not a failure to hide: the simple pilots are strong because they compute with the same physics, and the places where looking ahead pays are few and narrow.

6.3 Limitations

- **Model limits:** no terrain; in the synthetic scenarios obstacles are 2D rectangles with a fixed 15 dB loss (only the Kathmandu case has building heights, and there 94% of them are drawn from the tagged 6%); no backhaul; no interference from other cells; no bit-error rate; the turning-energy term is our own extension; walking speeds, surge behaviour and minimum rates are engineering assumptions; fading is off in the main runs; the shadowing and hidden-building values of the mismatch study are engineering assumptions.
- **Information:** most baselines know every person's true position. The value-of-information study shows that phone reports from about half of the phones would be enough, but it assumes phones report independently of where they are and that the share of reporters is known; locating phones that do *not* report is future work (Section 7.3).
- **NOMA practicality:** ordinary phones are believed not to support NOMA decoding (to be checked); 10-17% of adaptive NOMA pairs give the weak user less than half the power, which practical receivers may not accept (no clamp tested).
- **Regulation:** our altitude range (50-300 m) is above the 120 m limit that drone rules commonly set for routine flights [PENDING/VERIFY: cite India's Drone Rules 2021 and the DGCA airspace map]. The best pilot often uses the top of the range: in the Kathmandu case B6 flies at about 290 m, and in the still-crowd mismatch scenarios it chose the 300 m ceiling in 20 of 20. Results with a 120 m ceiling need a re-run.
- **Statistics:** several experiments use 12-20 scenarios per cell (the anticipation test 12, the hybrid and mismatch tests 20, the Kathmandu case 20); the sim-to-sim check uses 5.
- **Quadcopter energy:** the coverage results carry over to the quadcopter model, but its battery numbers are not yet trustworthy (Section 5.10).
- **Real place:** the Kathmandu people and flood band are reconstructed, not observed; the flooded localities rest partly on news reports.
- **Single drone:** H4 was not attempted.

6.4 Threats to validity

- **Internal:** a bug could favour one pilot. Mitigations: one shared `env.step()`, paired comparisons on identical scenarios, tests that new features leave old results bit-exact, a negative control (an AI must not beat B6 where the clairvoyant shows no room), and independent reviews that found real problems (an unfair OMA reference, the shaking exploit, the forecasting pilot leaving too early).
- **Construct:** coverage counts users at their minimum rate; it ignores how much more than the minimum they get (we report total rate and fairness alongside). The minimum rates are not signed off.
- **External:** the data are synthetic (and in the Kathmandu case the people are reconstructed on a real map); absolute coverage numbers do not transfer to a real disaster, only the comparisons and qualitative lessons do. The Kathmandu case supports this: the pilots rank the same on real geometry as on the synthetic map. The Al-Hourani parameters are from secondary sources; MATLAB parity (Gate G1) is not done.
- **Conclusion:** pass marks were written before results (G4, competing sites, G5, value of information, anticipation rule, hybrid timing, model mismatch) and the test seeds are untouched, so the verdicts cannot have been tuned to the outcome. Two departures are disclosed where they occur: the model-mismatch training cells were added after the pre-registered grid gave a null (exploratory), and the anticipation rule's candidate set was extended once before the new designs were run.

Chapter 7. Conclusion and future work

7.1 Conclusion

We built an open, tested simulation-based planning tool for a UAV base station serving disaster survivors with a NOMA downlink, and used it to test four hypotheses fairly.

Table 7.1 Hypotheses: verdicts

Hypothesis	Verdict	Evidence
H1 adaptive beats fixed	**Supported** where area is large and power scarce	`hard`: 0.741 vs 0.428 (Table 5.1); small gap in default scenarios (0.914 vs 0.882); Kathmandu case +0.107 (crowd near the centre, nobody walks)
H2 RL beats heuristics only under dynamics and battery limits	**Not supported**	G4 not met (0.576 < 0.580); G5 round 1 all agents below the bar (best 0.680 vs 0.756); hybrid timing AI -0.004 / -0.019 vs the bar; model mismatch -0.039 / -0.101; no room on walking crowds (+0.002); room confirmed only on 2 km competing sites
H3 NOMA beats OMA	**Supported**, with an honest trade-off	Adaptive NOMA +0.03 to +0.05 over equally optimised OMA for every pilot; +0.127 for B6 on real Kathmandu geometry (twice `hard`); +0.08 to +0.14 for every AI agent (G5, hybrid, mismatch); fixed NOMA loses coverage at 30 dBm (232 kbps ceiling); optimised schemes trade total rate for people served
H4 multi-UAV	**Not attempted**	Stretch goal

Further findings and contributions:

- **Energy:** circling at the minimum-power speed serves about 26% more people per battery (robust +21% to +31%); most of a search pilot's lead came from choosing the altitude.
- **Clairvoyant headroom:** a test that tells, before any training, whether a scenario leaves room for a learning controller; room was confirmed on fresh scenarios only where demand jumps between sites about 2 km apart. A shortcut formula for that room (the anticipation rule) was **not confirmed** on 8 new designs.
- **Value of location information (passed):** a planner keeps 98% of its coverage with position reports from half of the phones on moving crowds and one in twenty on competing sites, with up to 50 m GPS error; reports should be at most 30 s old.
- **Sim-to-sim:** a quadcopter with an autopilot and wind flies our plans within 0.012 coverage of the simplified game.
- **Real geometry:** on a scenario based on the 2024 Kathmandu floods, with 12,061 real buildings, the pilots rank as on the synthetic map and NOMA's gain doubles for the best pilot.
- **The map of where AI helps:** nowhere in our tests did learning beat computed rules; we can say why (inputs, announcement use, altitude, little room) and where it could (only where the clairvoyant test shows room).

7.2 Future work

1. **Final evaluation:** read the untouched test seeds (120-139) once for the main comparisons, with the negative controls; merge the four study branches into `main`.
2. **Find-and-serve:** locate survivors from their phones' uplink signals and serve them in the same mission (Section 7.3; study in progress).
3. **Real places, further:** population density for crowd placement, satellite flood extents instead of a fixed band, terrain elevation, a test of the geometric line of sight against a measurement; always labelled "a scenario based on ...", never real survivor data, since survivor positions are not published.

4. **Tethered drones:** a tether supplies power and backhaul, removing the battery limit and the backhaul gap, at the cost of a fixed ground station and a limited radius; drone cell sites of this kind (for example AT&T's "Flying COW" after Hurricane Maria in 2017 [36]) have been reported in news and operator sources [VERIFY].
5. **Legal altitude limits:** re-run the main comparisons, and the Kathmandu case where B6 flies at about 290 m, with the ceiling at 120 m above ground and the corresponding permission rules [VERIFY: Drone Rules 2021].
6. **The AI, if pursued further:** give it the altitude choice and a direct use of the announcement (the two largest parts of its shortfall, Table 5.13), and run the clairvoyant test first on any new scenario family, since the anticipation rule cannot replace it.
7. **Disaster events and realism:** rising flood, aftershocks, drone failure; wind, GPS error and command delay; a safety layer (geofence, battery reserve, return home, fallback to B6).
8. **NOMA as a decision of the AI**, need-aware pairing, bit-error rate, and a bench test with software-defined radios.
9. **Validation:** read the Al-Hourani paper, MATLAB parity, sign-off of the minimum rates and their sensitivity runs, and one real measurement of the signal against height.
10. **Multi-UAV control (H4)** and a flight in ArduPilot SITL over MAVLink.
11. **Business track:** the most realistic product for a student team is software only, a planning and rehearsal tool for emergency teams or telecom operators; the next step is to interview 5-10 practitioners before building more.

7.3 Future work: finding survivors from their phones (find-and-serve)

Status: study in progress; results pending. Nothing in this section is a result yet.

The idea in plain words. After a disaster the drone is the only cell tower in the area, so every working phone nearby tries to attach to it and transmits. The strength of a phone's uplink signal at the drone says roughly how far away the phone is: not a point, but a noisy ring around the drone. When the drone measures the same phone from several spots, the rings cross, like GPS working in reverse, and the phone's position can be estimated. Noise widens the rings: shadowing by buildings (correlated over tens of metres, which is why the Gudmundson model [34] is used), extra loss through rubble, and fast fading. The drone then has a trade-off: flying to new spots to sharpen the estimates (search) or staying where it serves the most people (serve), with NOMA sharing its radio among them. Phones transmit with limited power (a typical handset is power class 3, 23 dBm [37]), so the uplink, not the downlink, limits how far the drone can hear them. The value-of-information study (Section 5.11) already tells how good the estimates must be for planning: about half of the phones, 50 m accuracy, refreshed every 30 s.

Related work [VERIFY all]. SARDO [33] describes a drone carrying a light LTE base station that localises phones for search and rescue. LifeSeeker [35] is reported as a commercial drone-borne phone-locating system used by the Spanish Guardia Civil. Neither, as far as we know, plans the drone's flight to *serve* the located people with NOMA at the same time; this still has to be checked.

Table 7.2 Path to real deployment

Stage	What	Who and what is needed
1 Simulation	Find-and-serve in this simulator: uplink rings, noise, search vs serve, NOMA	This project (study in progress)
2 Lab test	A software-defined radio running an open-source LTE stack as the base station, real phones, in a shielded room or under a test licence	A university lab; radio test permission
3 Field trial	The base station on a drone over a mock disaster site	An operator or a rescue agency as partner; in India likely a WPC test licence for the spectrum and DGCA permission for the drone [to be checked]
4 Approval and adoption	Use in real responses	Telecom operators, regulators, and agencies such as the NDRF

Real-world barriers.

- **It is technically an IMSI-catcher:** a base station that makes phones attach. Using licensed spectrum needs a licence and the operator's cooperation; operating one without authorisation is generally restricted [to be checked for India].
- **Authentication:** a phone only completes the attach when the operator's core network authenticates its SIM. A standalone drone station without that link can only hear attach attempts, which may still be enough to measure signal strength, but not to give service.
- **Phone battery:** survivors' phones may be switched off or flat if rescue takes long.
- **Rubble and buildings** attenuate the uplink strongly and bias the distance rings.
- **The drone:** payload, endurance and airspace permissions limit what can be carried and for how long.
- **Competing proven tools:** thermal cameras, search dogs and acoustic listening devices are established in rescue work; a new tool must add something they cannot.
- **Cost and coordination** between operators, regulators and rescue agencies.

Results. [PENDING: results of the find-and-serve study, to be inserted when the study is finished and its pre-registered checks are run. Do not quote any number from this section until then.]

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[PENDING: every reference below must be checked against the original by a team member (see literature.md, all marked VERIFY). Journal volume and page details marked "verify" were recalled, not confirmed online.]

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